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Chapter 1

The Fundamentals of Gravitational Wave Interferometers

1.1 Introduction

On the 14th of September 2015, the detection of a gravitational wave emanating from two coalescing black holes signalled the beginning of a new era for astronomy [1]. The information carried by a gravitational wave is undisturbed by the matter between the observer and the astrophysical body that is emitting them, thus it gives astronomers valuable information which cannot be obtained from the electromagnetic radiation emitted by that body. Gravitational wave astronomy also allows for direct observations of objects that do not emit electromagnetic radiation, e.g. black holes. Therefore, it is important that large numbers of gravitational wave signals are measured to develop our understanding of the universe.

However, gravitational waves are incredibly weak and the detection of them is a science in itself. This chapter will outline how a gravitational wave interferometer, with a particular emphasis on LIGO [2], can be used to detect these signals and the fundamental noise sources these detectors face. While there are other important sources of noise, the suppression of quantum noise is crucial to achieving the sensitivity required to detect gravitational waves; therefore, quantum noise is the focus of this chapter.

1.2 The Astrophysical Origins of Gravitational Waves

Gravitational waves produced by inspiraling pairs of massive objects, such as binary black holes (BBH) (e.g. [1, 3–5]), binary neutron stars (BNS) [6, 7], and a source which is either a BBH or a binary consisting of a black hole and a neutron star (NSBH) [8], have been detected. Every confirmed gravitational wave signal to date is summarised in [9]. The waveform of the gravitational radiation emitted from these systems is known as a chirp because the frequency of the gravitational wave signal increases as the two objects get closer to merging [10]. The cut-off frequency is determined by the radius of the objects that will merge; for the signals which have been measured, the cut-off frequencies were between 100 Hz and 10 kHz. A typical gravitational wave strain amplitude of these types of systems, all of which have a total mass of $10M_{\odot} - 100M_{\odot}$ and are of the order 100 Mpc to 10 Gpc away from Earth, is $h \sim 10^{-21} - 10^{-23}$.

Einstein’s general theory of relativity accurately describes the gravitational waves measured with the LIGO and Virgo interferometers [11]. Gravitational waves are

emitted from mass quadrupole systems (see e.g. [12] or [13]). As the coupling between mass and the curvature of space-time is weak, gravitational waves are only observable in systems that have large masses moving at relativistic velocities.

Other astrophysical phenomena will emit gravitational waves. Asymmetries in pulsars [14,15] and magnetars [16] will cause them to emit a gravitational wave with a constant frequency. When the core of a star collapses as it goes supernova, a burst of gravitational radiation may be emitted [17,18]. Burst signals may also be generated immediately after a BNS merges [19]. Weaker sources that future gravitational wave interferometers may detect include: a stochastic background level of gravitational waves due to the merging of compact binaries from all over the universe [20], dark matter [21,22], and sources that existed during the early universe [23,24].

To improve statistical models of the systems that emit gravitational waves, the number of detections needs to be increased. To detect continuous wave signals, the signal to noise ratio (SNR) improves with the total observing time of the detector. With longer observation times, it is more likely that an exotic event will be measured. For these reasons, gravitational wave detectors are operated for year long time-scales with minimal interruption. These are known as observation runs. Between observation runs, the detectors undergo hardware upgrades. The previous, current and future observation runs are summarised in Table 1.1; a more comprehensive overview is found in [25].

Observation Run	Dates	Detections	Detectors
O1	12/09/2015 - 19/01/2016	3	aLIGO
O2	30/11/2016 - 25/08/2017	8	aLIGO, Virgo
O3a	01/04/2019 - 01/10/2019	39	aLIGO, Virgo
O3b	01/11/2019 - 30/04/2020	23	aLIGO, Virgo
O4	2021	-	aLIGO, Virgo, KAGRA
O5	2025	-	LIGO A+, Advanced Virgo, KAGRA, LIGO India

Table 1.1: Summary of observation runs [25] for LIGO, Virgo and KAGRA [2,26,27].

1.3 Measuring Gravitational Waves with an Interferometer

Gravitational waves are propagating longitudinal perturbations in the curvature of space-time, and can be split into a basis set containing two polarisations, plus and cross (e.g. [12] or [13]). The effect these waves have, in the local Lorentz frame, on free test masses on Earth is to move them by a small amount ¹.

To detect plus polarised gravitational waves, the end mirrors of a Michelson interferometer can be used as the test masses. These mirrors are referred to as *end test masses* (ETMs). A suspended mass will behave in the same way as a free mass when it is driven at above its resonant frequency, so the ETMs need to be

¹In the local Lorentz frame of the beam splitter, for any feasible Earth based detector, the effect of a gravitational wave on the wavelength of the light is negligible compared to its effect on the positions of the test masses due to the strong equivalence principle: All non-gravitational laws of physics which can be expressed in a special relativistic form have the same form in a local Lorentz frame (e.g. [12]). In the local Lorentz frame, the metric has the form $g_{\alpha\beta} = \eta_{\alpha\beta} + \mathcal{O}(h)$, where $\eta_{\alpha\beta}$ is the Minkowski space-time metric and $\mathcal{O}(h)$ is a perturbation of the order $\frac{4\pi L}{\lambda_{\text{EM}}}^2 h$. The Minkowski space-time metric is flat, so in the local Lorentz frame the metric is, essentially, flat and the gravitational waves cause the test masses to move by $\Delta L = hL$. Maxwell's equations are modified by a term that is of the order $\frac{\delta\lambda_{\text{EM}}}{\lambda_{\text{EM}}} \sim \frac{4\pi L}{\lambda_{\text{EM}}}^2 h$, and so if $\frac{4\pi L}{\lambda_{\text{EM}}}$ is small, as it would be for an Earth based interferometer, then the effect of the gravitational wave on the light is much smaller than the effect of the gravitational wave on the test masses.

suspended [28]. A sketch of a gravitational wave interacting with a Michelson interferometer is shown in Figure 1.1. For test masses at a distance L away from the origin of the local Lorentz frame, the distance they move, ΔL , due to an incoming plus polarised gravitational wave of strain h is

$$\Delta L = \frac{1}{2}Lh. \quad (1.1)$$

Interferometers are sensitive to arm length changes that are on the order of one wavelength, λ_{EM} , of the light they are operated with. As a plus polarised gravitational wave will move the test masses in anti-phase, the difference between the phase of light when it recombines at the beam splitter after reflecting from the test masses, $\Delta\phi$, is

$$\Delta\phi = \frac{4\pi}{\lambda_{\text{EM}}}Lh. \quad (1.2)$$

$\Delta\phi$ is known as the *differential phase*, and differential effects result in the light acquiring differential phase. The wavelength current interferometers operate with, $\lambda_{\text{EM}} = 1064 \text{ nm}$, was selected because test masses with low noise and mirror coatings with low loss and low thermal noise can be fabricated for this wavelength [29].

The gravitational wave strain is converted into intensity fluctuations of light at the *antisymmetric* port (see Figure 1.1) due to the two light fields from the arms interfering at the beam splitter. When on the *dark fringe*, there is a small amount of light at the antisymmetric port for the signal light to beat with and so the variation in the intensity of the light depends linearly on the gravitational wave strain. The interferometer is set to operate on the dark fringe by introducing a small static differential offset, ϕ_0 , in the differential arm length. This technique is known as DC

readout [30, 31]. At the beam splitter, the electric field, Ψ_{AS} , emerging from the antisymmetric port due to the interference of the beams returning from the ETMs, Ψ_y and Ψ_x , with differential phase $\phi = \phi_0 + \Delta\phi$ is

$$\Psi_{\text{AS}} = \frac{\Psi_y}{\sqrt{2}} - \frac{\Psi_x}{\sqrt{2}}, \quad (1.3)$$

$$= \sqrt{\frac{I_0}{2}} \left(\frac{e^{i(\phi_0 + \Delta\phi)}}{\sqrt{2}} - \frac{e^{-i(\phi_0 + \Delta\phi)}}{\sqrt{2}} \right), \quad (1.4)$$

where I_0 is the power of the beam injected into the beam splitter by the laser. The intensity of the beam at the antisymmetric port, I_{AS} , is

$$I_{\text{AS}} = \Psi_{\text{AS}}^* \Psi_{\text{AS}}, \quad (1.5)$$

$$\simeq I_0 \phi_0^2 + 2I_0 \phi_0 \Delta\phi, \quad (1.6)$$

$$= I_0 \phi_0^2 + 2I_0 \phi_0 \frac{4\pi}{\lambda_{\text{EM}}} Lh. \quad (1.7)$$

In the absence of a gravitational wave signal, the DC light level at the antisymmetric port is related to the static offset by $\phi_0 = \sqrt{\frac{\overline{I_{\text{AS}}}}{I_0}}$, where $\overline{I_{\text{AS}}}$ is the average of I_{AS} . As we are interested in a fluctuating signal, the DC offset from Equation (1.7) can be removed, allowing for the signal at the antisymmetric port to be written as

$$I_{\text{AS}} = 2\sqrt{I_0 \overline{I_{\text{AS}}}} \frac{4\pi}{\lambda_{\text{EM}}} Lh. \quad (1.8)$$

Therefore, it is possible to measure the differential motion of the ETMs due to a gravitational wave's strain by placing a photodiode in the beam emerging from the antisymmetric port of a Michelson interferometer.

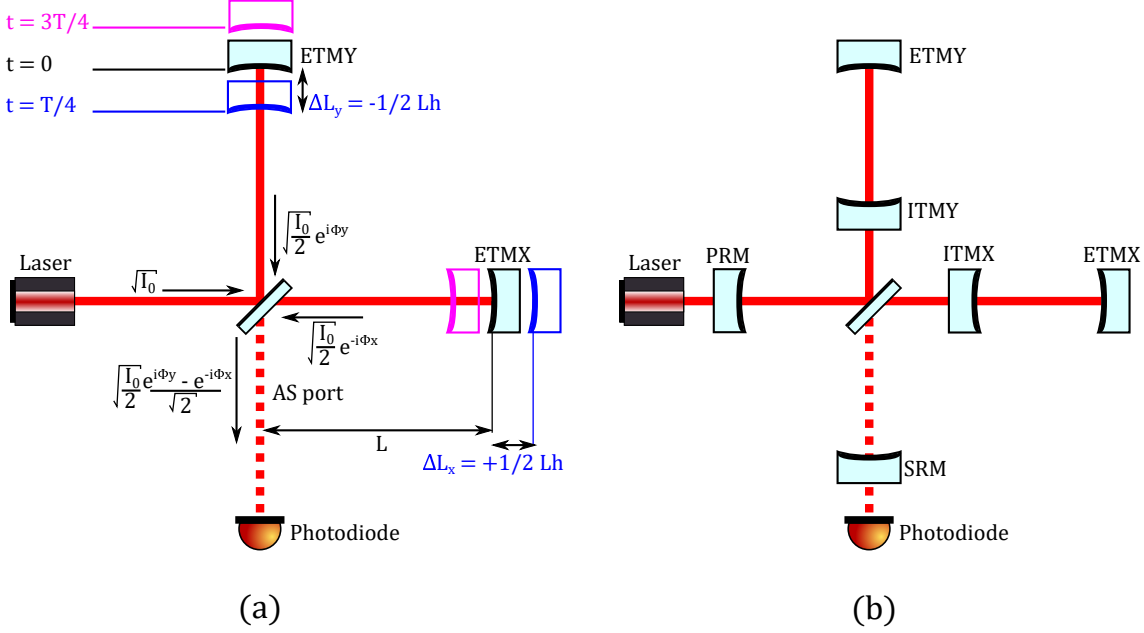


Figure 1.1: (a) A sketch of the effect a gravitational wave has on a Michelson interferometer. The positions of the test masses, ETMX and ETMY, at different times during the gravitational wave's period is shown by the blue and pink mirrors. The electric fields within the interferometer are indicated: $\sqrt{I_0}$ is the laser's amplitude and ϕ_x and ϕ_y is the phase acquired by the light as it travels up and down the arm. (b) To reach the sensitivity needed to detect gravitational wave, several mirrors need to be added to the basic Michelson interferometer: ITMX and ITMY boost the signal generated in the arms, the PRM increases the power at the beam splitter, and the SRM can be used to shape the quantum noise of the interferometer.

1.3.1 Using Fabry-Perot Cavities in the Arms of a Michelson Interferometer to Enhance Its Sensitivity

The optimal arm length of the interferometer is determined by $1/4$ of the wavelength of a gravitational wave as this gives the maximal $\Delta\phi$. Gravitational wave signals from BBH and BNS have frequencies of ~ 100 Hz; therefore, the wavelength of the gravitational wave will be of the order ~ 3000 km. It is not possible to construct a Michelson interferometer with ~ 1000 km arms on Earth.

The sites that were available to construct the LIGO detectors allow for 4 km arms. To reach the optimal arm length on a 4 km site, the number of times that the light bounces off of the ETMs, B , needs to be increased. Numerically, B should be ~ 250 . This means that Equation (1.2) becomes

$$\Delta\phi = \frac{4\pi}{\lambda_{\text{EM}}} BLh. \quad (1.9)$$

Fabry-Perot cavities are used to increase the number of times light bounces off of the ETMs². Cavities are formed by placing an *initial test mass* (ITM) in each arm near the beam splitter. When a photon enters a cavity, it remains in the cavity for an average number of bounces that is determined by the reflectivity of the mirrors. Interferometers measure $\Delta\phi_{\text{EM}} = \frac{d\phi_{\text{EM}}}{d\phi_{\text{ETM}}} \Delta\phi_{\text{ETM}}$, so when a Fabry-Perot cavity is near or on resonance, the Fabry-Perot cavity enhances the detectors sensitivity as the slope $\frac{d\phi_{\text{EM}}}{d\phi_{\text{ETM}}}$ is much greater for a Fabry-Perot than it is for a mirror. This is

²Herriott delay lines were considered as a means of achieving more bounces from the ETMs, however due to difficulty in manufacturing suitable mirrors which do not scatter light, thus generating noise, delay line interferometers fell out of favour. In principle, both delay lines and Fabry-Perot cavities would perform equally well due to the Mizuno limit [32].

illustrated in Figure 1.2. The exact way this happens is dependent of the frequency of the gravitational wave and the reflectivity of the mirrors, with the arm cavity acting as a low pass filter (e.g. see [32, Chapter 3], [33, Appendix D], [34] or [35]).

1.4 Sources of Noise in Gravitational Wave Interferometers.

As the strain induced by a gravitational wave causes a minuscule change in the phase accumulated by light in the arm cavities, the challenge of detecting gravitational waves lies in suppressing the noise below this signal. From Equation (1.9), for a gravitational wave strain of $h_{\text{rms}} = 10^{-23}$ interacting with LIGO, the beams interfering at the beam splitter will have a difference in phase of $\sim 10^{-11}$ rad.

1.4.1 Laser Power and Shot Noise

The shot noise associated with detecting a coherent state, such as that created by the laser, can limit the sensitivity of the interferometer. To reduce the level of shot noise, the energy, E , of the laser needs to be high. The uncertainty in energy, ΔE , that associated with a measurement of the time between two events, ΔT , is determined by the time-energy uncertainty principle, $\Delta E \Delta T \geq \hbar/2$. The uncertainty in the phase, $\Delta \phi$, of a light beam with angular frequency ω_{EM} is related to the measurement time by $\Delta \phi = \omega_{EM} \Delta T$ [36]. The phase is averaged over a time corresponding to half the gravitational wave's period, so $\Delta T = \frac{1}{2f_{\text{GW}}}$. The energy used in the measurement is $E = I_0 \frac{1}{2f_{\text{GW}}} = \hbar \omega_{EM} N$, where I_0 is the power of the laser and N is the number of photons detected in the averaging time. As the

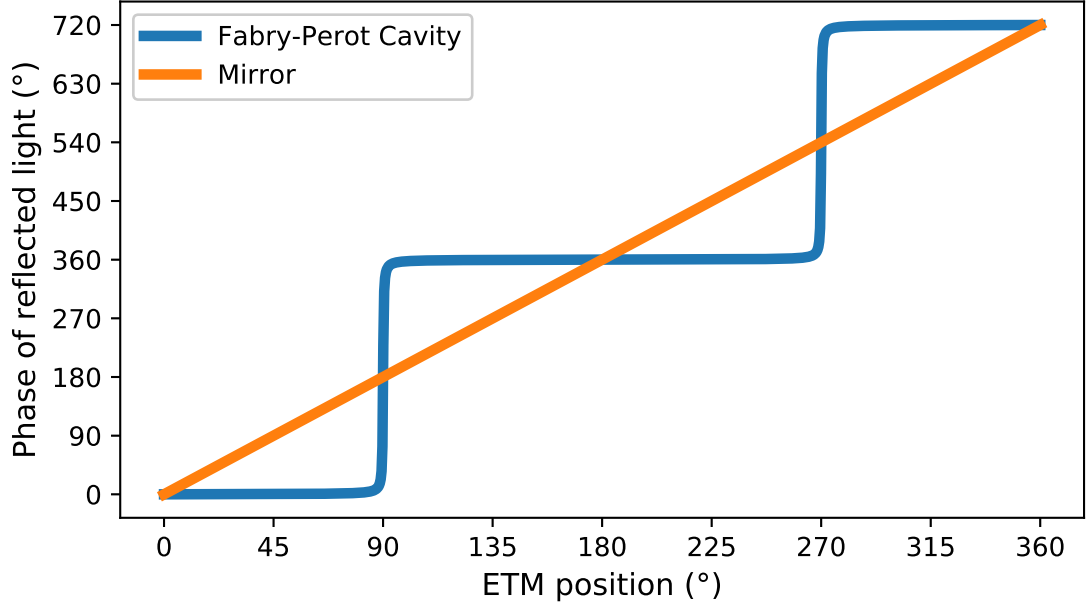


Figure 1.2: Comparison between the phase of the light reflected from a mirror versus the phase of light reflected from a cavity. The blue line shows the phase of the light that is reflected by a Fabry-Perot cavity. Initially, the cavity is on anti-resonance so perfect destructive interference occurs as the round-trip phase is 180° . The end mirror (ETM) is moved 90° , resulting in an extra 180° round-trip phase, and so the cavity resonates. The phase of the light reflected from the cavity rapidly changes as it moves through the resonance. The ETM is moved further, the cavity stops being resonant and the phase of the light is not changed as its reflected by the cavity. The orange line shows the phase of the light when a mirror is used instead of a cavity. The interferometers sensitivity is proportional to the slop $\frac{d\phi_{EM}}{d\phi_{ETM}}$, and so by inserting another mirror in the arm near the beam splitter, a Fabry-Perot cavity is formed and the sensitivity of the interferometer is increased.

laser is in a coherent state (e.g. [37–39]), the uncertainty detecting N photons is $\Delta N = \sqrt{N}$; therefore, the uncertainty in the energy used in the measurement is $\Delta E = \hbar\omega_{EM}\sqrt{N}$. Using the expressions for ΔE and $\Delta\phi$, the uncertainty principle may be used to obtain

$$N \geq \frac{1}{4(\Delta\phi)^2}. \quad (1.10)$$

The amplitude spectrum of the intensity measured by the photodiode over a time corresponding to a bandwidth of Δf at the antisymmetric port will depend on the amount of light at the dark port,

$$I_{\text{ASrms}} = \sqrt{2\overline{I_{\text{AS}}}\hbar\omega_{EM}\Delta f}. \quad (1.11)$$

The conversion from light power at the antisymmetric port to gravitational wave strain can be inferred from Equation (1.8),

$$\frac{h}{I_{\text{AS}}} = \frac{1}{2\sqrt{I_0\overline{I_{\text{AS}}}}}\frac{\lambda_{\text{EM}}}{4\pi LB}, \quad (1.12)$$

and so the amplitude spectrum of the strain corresponding to the shot noise of the light at the AS port is

$$I_{\text{ASrms}} = \sqrt{\frac{\hbar c^2}{8I_0\omega_{EM}L^2B^2}\Delta f}. \quad (1.13)$$

Equation 1.13 assumed the frequency response of the interferometer to a gravitational wave is flat, however the arm cavities have a low pass filtering effect on the signal, and so the shot noise will rise as f above the characteristic frequency of the arm cavity. This can be seen in the spectral density of the LIGO detector’s noise (Figure 1.3). Equation (1.13) is more rigorously derived in [34].

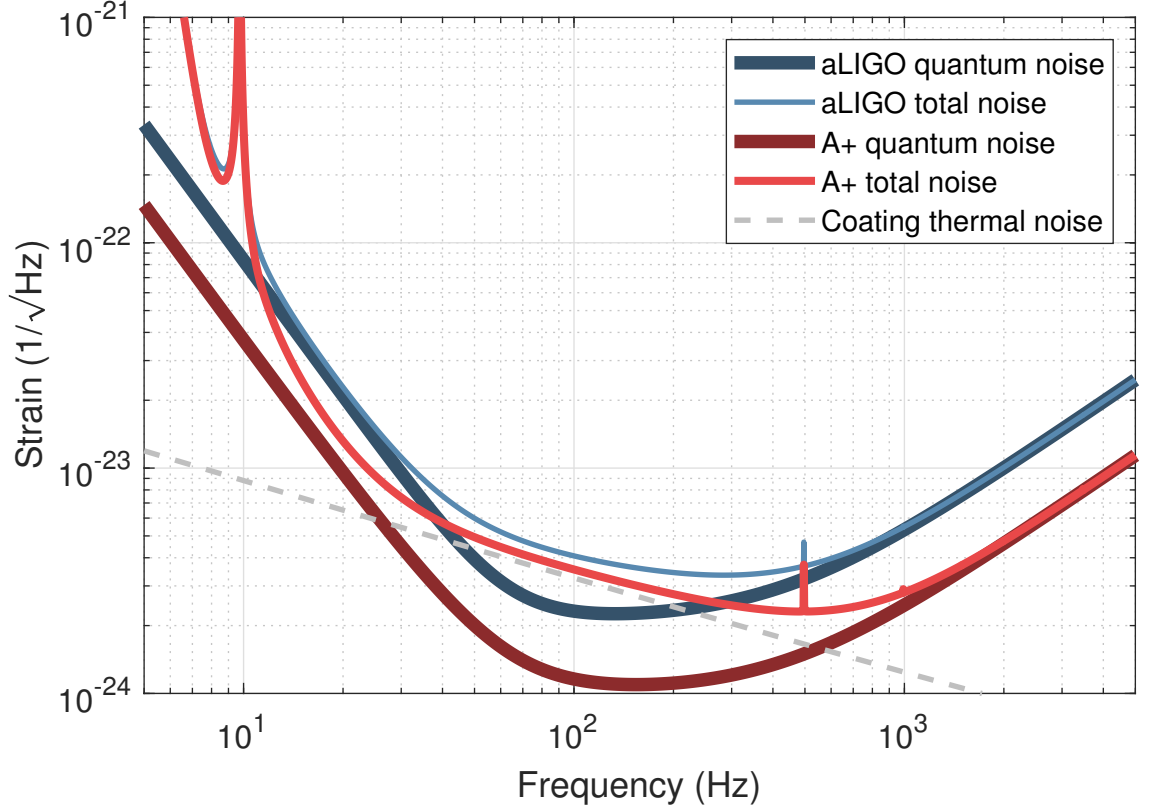


Figure 1.3: The strain Sensitivity of the aLIGO and A+ detectors. The noise due to radiation pressure and photon shot noise is shown in dark-blue (aLIGO) and dark-red (A+). Above 10 Hz, quantum effects are one of the limiting sources of noise; with the other being the noise due to Brownian motion of the mirror's coating. Below 10 Hz, seismic noise begins to dominate the noise of the detector. Above 100 Hz, shot noise dominates the noise of the detector. This rises with f as the shot noise is filtered by the arm cavities. Below 100 Hz, the quantum noise goes as $1/f$ due to radiation pressure.

To beat the photon shot noise by a factor of 10, Equation (1.9) and Equation (1.10) show that, for a LIGO-like interferometer, the required number of photons to measure a gravitational wave with a strain amplitude of $h_{\text{rms}} = 10^{-23}$ is $\sim 10^{22}$. For a gravitational wave signal with a frequency of 100 Hz, the laser power incident on the beam splitter would need to be $\sim 1\text{MW}$. No such laser exists.

1.4.2 Using Power Recycling to Reduce Shot Noise

To increase the laser power that is incident upon the beam splitter, a power recycling cavity can be used [40–42]. When the interferometer is on the dark fringe, most of the light from the laser gets reflected by it. Thus, the interferometer is, effectively, a mirror. Drever [43] [can't find a copy!](#) proposed that a *power recycling mirror* (PRM) could be placed between the laser and the interferometer to form a cavity. A sketch of this is shown in Figure 1.1. The total power impinging on the beam splitter would be increased by making wavelength scale position changes of, i.e. *tuning*, the position of the PRM such that the light reflected from the interferometer and the light from the laser add in phase, i.e. the reflected light power would be ‘recycled’. Current laser sources that can be operated for year-long timescales are capable of producing of the order 100W, see Chapter ??, and so the power recycling factor needs to be $\sim 10^4$. This technique has been implemented at LIGO [44] and other large-scale interferometers [26, 27].

1.4.3 Using Signal Recycling to Shape Quantum noise

The interferometer’s bandwidth depends on the amount of time that signal-carrying light is in the arms for. Meers suggested that the signal light could be recycled in

a similar fashion to how power is recycled [40]. This detection bandwidth can be altered by introducing a *signal recycling mirror* (SRM). A coupled cavity, known as the *signal recycling cavity* (SRC), between the SRM, ITM and ETM will be formed by placing the SRM between the photodiode and the beam splitter. A sketch of this is shown in Figure 1.1. The finesse of the arm cavities as seen by the light emerging at the antisymmetric port depends on the tuning of the SRM. There are two common modes of operation: *resonant sideband extraction* (RSE) [45, 46] and *signal recycling*. Alternatively, the SRM can be tuned so that the interferometer is optimised for a narrow band of frequencies to increase its sensitivity to certain BNS signals, e.g. [47].

When in RSE mode the SRM is tuned so that the SRC is anti-resonant, thus broadening the interferometers sensitivity bucket at the expense of losing peak sensitivity. This was motivated by needing to reduce the thermal load on the beam splitter by increasing the finesse of the arm cavities [46]. However, by increasing the finesse of the arm cavity, the bandwidth of the interferometer is reduced. To retain the desired arm cavity finesse from a sensitivity point of view, the SRM can be tuned so that the finesse of the arm cavity experienced by the signal is lowered.

In signal recycling mode, the SRM is tuned so that the SRC is resonant, deepening the sensitivity bucket at the expense of bandwidth. The trade-off between bandwidth and peak sensitivity comes from the conservation of energy. From an energy argument, the sensitivity of a detector to gravitational waves is determined by the detectors bandwidth and the energy within this bandwidth; so for the same amount of power at the beam splitter, this can be thought of as a fundamental limit

to the interferometer’s sensitivity. This is known as the Mizuno limit [32].

1.4.4 Reducing Shot Noise by Removing Junk Light with a Mode Cleaner Cavity.

All light that is incident on the output photodiode contributes to the total shot noise of the measurement; therefore, only light which carries signal should be incident on the photodiode to enhance the SNR of the measurement. In an ideal interferometer, the light emerging at the AS port will be a beam that is defined by the mode that is resonant within the arm cavity. To minimise diffraction losses and due to the mirrors’s manufacturing tolerances, the light in the arm cavity will be in the *fundamental Gaussian mode*. However, due to optical imperfections, e.g. astigmatism, and the RF side-bands used to control the interferometer, the light emerging from the AS port contains light in higher order modes. This undesired light is referred to as *junk light*.

To remove junk light, mode cleaners are used (e.g. [48, 49]). A mode cleaner is a high finesse cavity that strips away higher order mode light and the control sidebands. As higher order modes acquire a different amount of phase as they propagate compared to the fundamental Gaussian mode, an optical cavity can be designed such that the fundamental mode resonates while the higher order modes do not. These higher order modes will be suppressed by a factor of the cavity’s finesse. However, it is possible for some higher order modes to not be suppressed if their round trip Gouy phase is nearly the same as that of the fundamental mode; thus, the cavity is designed so that only higher order modes which carry little light will do this. The control sidebands are attenuated if they are outwith the cavities

linewidth. The beam transmitted by the mode cleaner will be predominantly in the fundamental mode, and so the SNR of the signal measured by the photodiode will be increased.

1.4.5 Radiation Pressure Noise and the Standard Quantum Limit

Radiation pressure noise is a quantum effect that arises due to the uncertainty in the momentum that light imparts to the test masses when it is reflected from them [37]. These quantum fluctuations will give rise to a differential force on the ETMs, meaning that this noise couples as $1/f^2$ to the measurement of the positions of the ETMs. As the force exerted by a beam is $F = \hbar\omega_{EM}N/c = P/c$, the amplitude spectral density due to the radiation pressure of the length of a Fabry-Perot cavity with mirrors of mass m is

$$\Delta x_{\text{RPNFP}} = \frac{\sqrt{\hbar\omega_{EM}P}}{2mc\pi^2 f^2}. \quad (1.14)$$

To reduce the effect of radiation pressure noise, the LIGO test masses are 40 kg. The effect of radiation pressure noise on the LIGO detectors's sensitivity curves can be seen in Figure 1.3.

Radiation pressure noise scales with the quantum fluctuations entering the anti-symmetric port whereas shot noise is inversely proportional to these fluctuations. For a laser which produces a coherent state, i.e. a classical laser, the standard quantum limit (SQL) [37, 50, 51] sets an optimal laser power for a given ETM mass which minimises the total quantum noise. The SQL results in a $1/f$ locus that is

determined by the mass of the ETM and the duration of the measurement.

The sensitivity curve of an interferometer with a signal recycling mirror that is tuned to either signal recycling mode or RSE mode will ‘touch’ the SQL at one frequency; however, for an arbitrary detuning of the SRM, the sensitivity curve of the interferometer will drop below the SQL at some frequencies due to non-linear optical effects [52].

1.4.6 Coating Thermal Noise and Other Noise Sources

There are many classical sources of noise to overcome in a gravitational wave interferometer. The most notable of these is the noise associated with the Brownian motion of atoms in the amorphous mirror coating layers of the test masses [53]. Current detectors are designed around the expected noise level that can be achieved with these coatings. This will be one of the limiting sources of noise in the A+ detector, as shown in Figure 1.3. Other noise sources include seismic noise from the motion of the Earth, Newtonian noise from pressure waves in the Earth’s crust, residual gas pressure noise that alters the refractive index of the arms, and the noise associated with the interferometer’s electronics (e.g. [54, 55]).

1.5 Reducing Shot Noise with Non-Classical States of Light

The two-photon formalism developed by Caves and Schumaker [56, 57] can be used to see how non-classical states of light can improve the sensitivity of an interferometer. In this framework, light is analogous to a quantum simple harmonic oscillator (SHO).

A SHO can be described by two orthogonal components, such as the momentum and position quadrature; for light a convenient basis to use is the *amplitude* and *phase* quadrature. The amplitude and phase quadratures, $\hat{\mathbf{X}}$ and $\hat{\mathbf{Y}}$ respectively, defined in terms of annihilation and creation operators of the electromagnetic field, $\hat{a}$ and $\hat{a}^\dagger$, are (N.B repetition with BHD chapter)

$$\hat{\mathbf{X}}(t) = \frac{1}{2} (\hat{a}(t) + \hat{a}^\dagger(t)) , \quad (1.15)$$

$$\hat{\mathbf{Y}}(t) = \frac{1}{2i} (\hat{a}(t) - \hat{a}^\dagger(t)) . \quad (1.16)$$

From this formalism, an uncertainty relation between the phase and amplitude quadratures can be derived:

$$\Delta\hat{\mathbf{X}}\Delta\hat{\mathbf{Y}} \geq \frac{1}{2} \quad (1.17)$$

If the uncertainty in one quadrature is reduced, the uncertainty in the other is increased. Such a light field is *squeezed*. The phase between the squeezed field and the detected quadrature is defined as the squeezing angle.

As the gravitational wave signals that are to be measured will be in the phase quadrature of the light, it would be advantageous to, by the uncertainty principle, move noise to the undetected amplitude quadrature from the detected phase quadrature; this can be achieved by using squeezed light. Squeezed light is created with an optical parametric amplifier; this is a device which generates correlated photons by converting single higher energy photons into two lower energy photons. This squeezed vacuum light is injected into the interferometer from the AS port so that the observed noise within the interferometer is squeezed. This technique has been implemented at the LIGO Livingston Observatory (LLO) [58] and GEO600 [59].

However, fluctuations in the amplitude quadrature of the light perturb the position of the test masses; thus, they couple to the signal at the antisymmetric port. During LIGO’s third observation run, the squeezing angle had to be tuned to prevent radiation pressure noise from limiting the BNS range of the detector [58].

To infer the quantum state of a light field, a technique known as *balanced homodyne detection* (BHD) can be used to observe one quadrature of the light field [60]. By combining the signal light field that is to be measured with a much stronger *local oscillator* light field on a beam splitter, the quantum state of the signal light field is encoded in the combined beam’s intensity. The relative phase between the signal and local oscillator determines the quadrature of the quantum state that will be observed. Chapter ?? explores this in more detail.

1.6 Interferometry Techniques to Beat Quantum Limits

Squeezed light and an effect known as ponderomotive squeezing [50], can be used to surpass the SQL at some frequencies. Ponderomotive squeezing arises due to the natural correlations between the phase and amplitude quadratures of the light created by the interferometer. A demonstration of the SQL being surpassed at the LIGO Livingston detector is found in [61]. Although [61] shows the SQL was beaten at some frequencies, the noise at other frequencies was increased relative to if no squeezing was used. As the LIGO detectors are optimised for sensing gravitational waves, beating the SQL is not the main objective when devising interferometer configurations. However, [61] shows that the 40 kg test masses do behave quantum-

mechanically. Additionally, sub-SQL interferometer configurations are optimised for detecting certain astrophysical sources with merging frequencies around the frequency at which the SQL is beaten.

To avoid the increase in radiation pressure noise that is experienced when squeezed light is used, a frequency-dependent squeezer can be used [50]. The light's squeezing angle can be rotated as a function of frequency. This is done by using a high finesse detuned filter cavity. At frequencies below the linewidth of the cavity, the light reflected from the filter cavity will experience a phase shift and so the squeezing angle will be rotated; in contrast, at frequencies above the cavity's linewidth, the light incident upon the cavity is reflected without experiencing any phase shift and so the squeezing angle is not rotated. At low frequencies, the amplitude quadrature can be squeezed and the radiation pressure noise can be decreased. At high frequency, the phase quadrature can be squeezed and the shot noise can be decreased. A Frequency dependent squeezed light source that is suitable for LIGO has demonstrated [62] and frequency dependent squeezing will be part of the A+ detector.

Variational readout is a technique where quantum noise due to radiation pressure is removed from the measurement by using a frequency dependent homodyne angle [50]. This is realised in a similar way to frequency dependent squeezing; the AS beam is coupled to a filter cavity and thus the detected quadrature of the beam emerging from this cavity is frequency dependent. This beam is then measured on photodiodes using a conventional BHD scheme. Variational readout schemes are sensitive to small amounts of loss within the filter cavity, and so this technology is not mature enough to be implemented into current gravitational wave detectors.

To beat the bandwidth limit, energy has to be injected into the system. White light cavities as a means of increasing the detector bandwidth have been studied [63–65]. However, these interferometer configurations have yet to be experimentally demonstrated.

There are several sub-SQL schemes which avoid the quantum noise problems related to making a position measurement of the test masses. If a momentum measurement of the test masses could be made, then a random impartation of ‘position’ would not affect future measurements of the momentum [66]. Such measurements are known as quantum non-demolition. As a speed measurement is being made, the displacement noise would rise as $1/f$ instead of $1/f^2$; this type of measurement can be made with a speed meter interferometer [67–69].

1.7 The Effect Losses Have on Measuring Quantum States of Light

To preserve the benefits gained by using quantum states of light, it is essential to have minimal losses in the read-out of a gravitational wave detector. Real optical components have physical attributes that result in incident light power being wasted. These mechanisms are known as losses, and examples of these include absorption and scatter. A lossy optical component can be modelled as an ideal one downstream of a beam splitter whose splitting ratio represents the loss (e.g. [38, Section 2]). The incoming light enters side of the beam splitter, some light is directed to the ideal component and some gets lost. In the remaining input to the beam splitter, vacuum fluctuations can enter. Some of these fluctuations will make it to the ideal

component, and so the state of the light that is detected is contaminated with extra vacuum fluctuations compared to if there was no loss.

Therefore, it is critical that the photodiode used to measure the AS beam has low loss. Photodiodes convert photons into electrons, ideally with a one-to-one ratio. The measure of how well this is done is the photodiode's *quantum efficiency*, η . The factor by which the shot noise increases, ϵ , due to a simple loss mechanism, such as a non-ideal photodiode, is

$$\epsilon = \sqrt{\frac{1}{\eta} - 1}. \quad (1.18)$$

The smallest photodiode quantum efficiency acceptable in a gravitational wave interferometer is 99% [70]. This can be achieved by photodiodes fabricated from InGaAs.

Mode matching losses have a greater contribution to the total shot noise compared to losses of the same size generated by a simpler mechanism [71]. This source of loss can be reduced through careful design of the optical layout. Mirrors with a tunable radius of curvature can be used to ensure that there is minimal mode matching loss. This is the focus of Chapter ??.

1.8 Overview of the LIGO Detector

The LIGO detector is a dual-recycled-Fabry-Perot Michelson interferometer with 4 km arms [2]. The layout of the detector is shown in Figure 1.4 [72] and its sensitivity curve is shown in Figure 1.3 [73, 74]. The term dual-recycled refers to the fact that the interferometer uses both power and, the mathematical dual of power

recycling, signal recycling. The interferometer is housed within a vacuum system comprised of several horizontal access modules (HAM) and beam splitter chambers (BSC). Figure 1.4 includes all the optics from the pre-stabilised-laser (PSL) to the readout photodiodes that the main beam interacts with.

The first cavity after the PSL is the input mode cleaner; this provides a beam with high spatial purity to the interferometer and is used in the initial locking sequence as a stable reference frequency. The power recycling cavity features two extra mirrors, known as PR2 and PR3, in addition to the PRM. The role of these mirrors is to make round trip gouy phase sufficient so that it is stable; the signal recycling cavity also has three mirrors for the same reason [44]. The beam passes through the output Faraday isolator (OFI), and is steered into the output mode cleaner. The gravitational signal is encoded in the beam transmitted by the OMC, so photodiodes monitor the light transmitted by the OMC.

1.9 The Future of Ground Based Gravitational Wave Detectors

The next generation of ground-based gravitational wave detectors include the Einstein Telescope [75], LIGO Voyager [76] and LIGO Cosmic Explorer [77]. LIGO Voyager will be a cryogenic interferometer installed in the current LIGO infrastructure and will be a factor of ~ 2 more sensitive than LIGO A+. The Einstein telescope Cosmic explorer interferometers will be ten times larger than the current generation of interferometers and will be able to test the no-hair theorem, test alternative theories of gravity, test models of string theory, measure objects with large

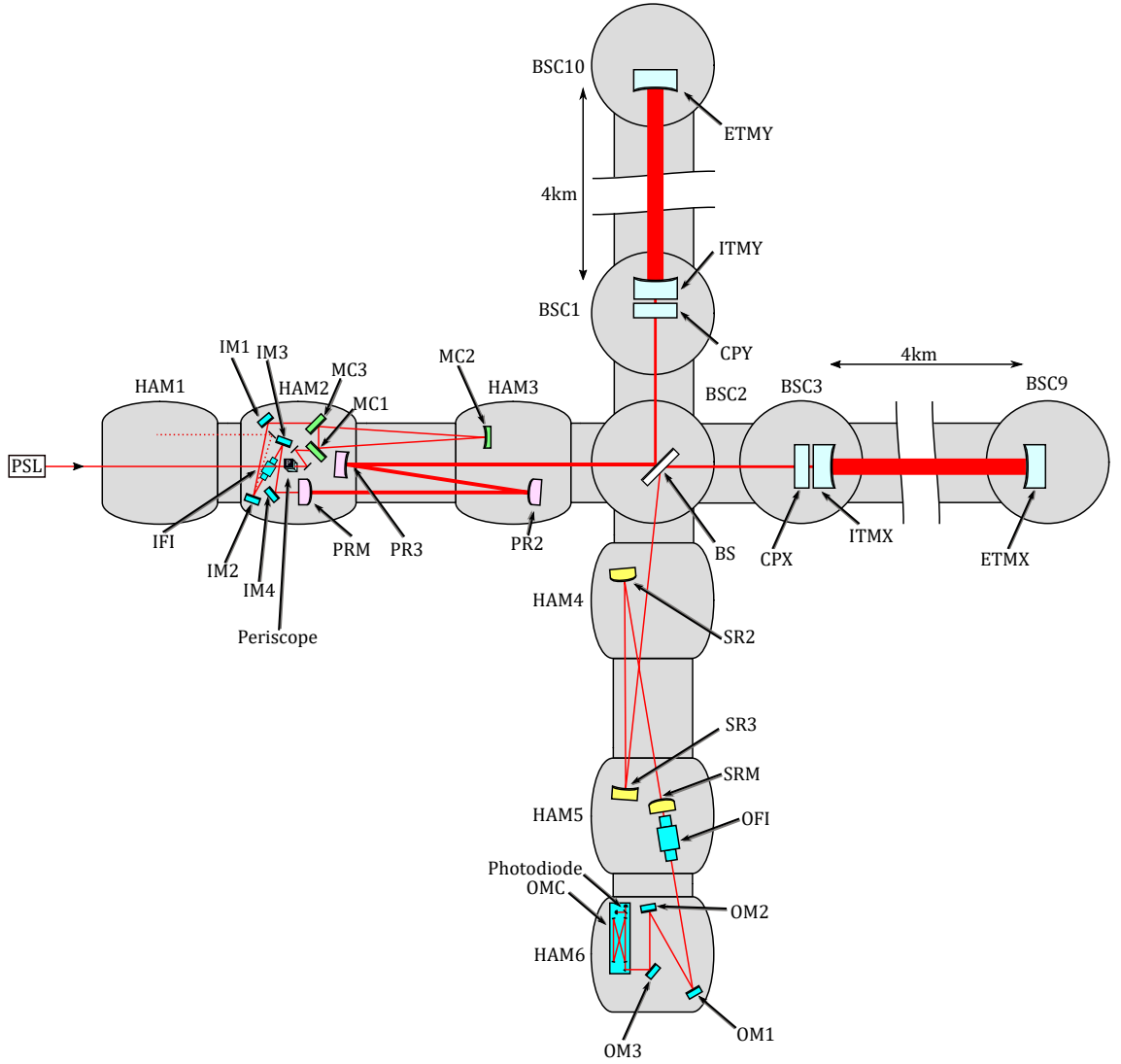


Figure 1.4: Sketch of a LIGO detector. The input and output optics, IM1-4, input periscope, OM1-3, OMC, IFI and OFI, are shown in blue. The input mode cleaner is shown in green. The power recycling is shown in pink. The arm cavities are shown in pale blue. The signal recycling cavity is shown in yellow.

redshifts to learn about the formation of cosmological structures, measure closer to the big bang than is possible with electromagnetic telescopes, detect cosmic strings, probe the physics of neutron stars and measure what happens during a supernova. As well as this, the third generation of gravitational wave detectors may measure signals from exotic sources which have not yet been theorised.

As the limiting sources of noise in current gravitational wave detectors is the coating thermal noise, research into cryogenic mirror coatings is under way [78]. Fused silica is opaque at these frequencies and its noise characteristics do not improve when cooled, so silicon test masses are being developed [76].

It is expected that low noise, low loss coatings will not be compatible with 1 μm light, so these detectors will operate with either a 1.5 μm or 2 μm wavelength laser [76]. Laser technology for 2 μm light is still in its infancy, and so there is lots of uncertainty in the practicality of using 2 μm as the operating wavelength of a gravitational wave detector; for instance, low noise, high quantum efficiency photodiodes that can detect 2 μm light have never been fabricated. This is discussed in Chapter ??.

1.10 Conclusion

Further detections of gravitational waves will lead to a better understanding of the universe. To reach the design sensitivity of the LIGO detectors, among other upgrades, the input laser power needs increasing. To further lower the LIGO detectors noise floor, the A+ upgrade will utilise frequency dependent squeezed light. For this to be successfully implemented, the quantum state of the light field must be

accurately detected with a balanced homodyne readout and losses in the read-out must be minimal.

As there is potential to reduce the noise floor that will limit the A+ detector by using cryogenic mirror coatings that are compatible with $2\text{ }\mu\text{m}$ light, future detectors may use a $2\text{ }\mu\text{m}$ laser; this leads to the question: is laser technology at this wavelength sufficiently mature enough to enable quantum limited measurements? Balanced homodyne detection and photodiodes with high quantum efficiency and low dark noise will be required for any future interferometer.

Chapter 2

The Consequence of Radius of Curvature Errors within the SRC on Modes at the SRM.

2.1 Introduction

A gravitational wave interacting with the LIGO detectors creates a differential signal in the arm cavities, and as the signal is differential it is directed to the Signal Recycling Cavity (SRC). The SRC tuning shapes the frequency response of a differential arm length change to light power transmitted through the antisymmetric (AS) port, with there being two modes of operation: signal recycling (SR) and resonant sideband extraction (RSE). The beam is then directed to the Output Mode Cleaner (OMC) to filter away ‘junk’ light. Junk light is comprised of higher order modes (both spatial and temporal), and some of this junk light comes from mode mismatches between the arm cavities and the SRC. The light transmitted by the

OMC ideally will only contain the signal generated by the differential arm length change, which is a signal that originated in the arm cavities. This will be explained in the ‘detector’ chapter, so does this need references?

Losses between the arm cavities and the readout photo-diodes result in less signal as well as more noise. The decrease in signal is straight forward to understand: if there are less photons to measure, then the signal will be smaller. Since the power is lower, the signal to noise ratio will be larger since shot noise goes as $\sqrt{\text{Power}}$. As gravitational wave detectors implement squeezing to decrease the quantum noise in the detection quadrature, it is important to preserve the quantum state of the signal field. This means that losses contribute additional noise, as losses pollute the squeezed quantum state with unsqueezed vacuum. This type of loss will be featured earlier in the thesis in the ‘detector’ chapter, so does this need references?

It is known that 20% of the light is lost between the AS port and the readout photo-diodes, as the shot noise can be compared to the expected shot noise for a given fringe slope, input power and known losses. Roughly half of the loss is accounted for e.g. some comes from the Faraday isolator, some comes from the OM3 pick off, however the other half is not fully understood e.g. see [79]. The level of squeezing can also be used to cross check the loss [80]. Some of the unaccounted for loss may come from mode mismatch between the arm cavities and the OMC.

Mode matching losses between the squeezed light source, the interferometer and the OMC act coherently. Mode mismatch can be modelled as a modal beam splitter that transfers some light from the HG00 mode to the HG20+02 mode. The amount of light transferred between the HG00 and HG20+02 mode is determined

by the magnitude of the mismatch, and the phase difference between the HG00 and HG20+02 mode depends on the nature of the mismatch. An error in waist size is in-phase (real), whereas an error in waist location is out-of-phase (imaginary). In general the mismatch introduces a complex phase. A sequence of mode mismatches can be modelled in the same way as a Mach-Zehnder interferometer. The difference in phase between the HG00 and HG20+02 mode is analogous to a difference in path length in a Mach-Zehnder interferometer. It differs from a typical Mach-Zehnder interferometer as the HG00 to HG20+02 splitting ratio is not well balanced. As the mode matching losses act coherently, the degradation of the squeezed state can be greater for mode matching losses than for other types of loss e.g. the loss due to photo-diodes with a quantum efficiency less than one. If the squeezer is perfectly matched to the OMC, then the mode mismatch between the squeezer and the interferometer is four times the loss you would expect from treating a mismatch loss as a regular loss. This is explained in detail in [81].

Since the SNR of the gravitational wave signal is directly related to mode-matching between the arm cavities and the OMC, knowledge of the beam parameter at the SRM is highly important. The mode in the arm is well defined since the arm cavities are high finesse. Uncertainties in the RoC of optics in the SRC telescope (SR2 and SR3) results in an uncertainty of the beam parameter of the arm mode at the SRM, as well as the eigenmode of the SRC. A fair amount of mode mismatch can be tolerated by the SRC as it is a low finesse cavity, so this leads to an uncertainty in the beam parameter that is transmitted through the SRM.

Junk light will be generated if the arms are perfectly matched to the OMC but not to the SRC. If the arms are matched to the OMC but not to the SRC, then actuating on the beam parameter with the ITM thermal lens and the SRM thermal lens would not increase the signal as these two actuators are separated by enough Gouy phase that a large portion of the beam parameter phase space can be covered with them. This means the mode matching between the arm and the OMC can be optimal but the light can still contain HOM. Additionally, if the beam is astigmatic, then optimal mode matching would not correspond to perfect mode matching.

The aim of this work is to see how the errors in the focal length of SR3 and SR2 effect the beam parameter of the arm mode at the SRM and higher order mode (HOM) content of the beam transmitted through the SRM. A LIGO-like interferometer was simulated using FINESSE. The simulated interferometer has properties very close to the the as built LLO LIGO detector, however some simplifications were made to make the effect that a geometry error in the SRC has on the beam transmitted through the SRM clearer. The difference in the HOM content of the beam transmitted through the SRM between an ITM single bounce measurement and a measurement which involves the arm cavities was investigated. The effect that the interferometer being in signal recycling and resonant sideband extraction (RSE) mode had on the HOM content of the beam as the focal length of SR3 deviated from its nominal value was also simulated.

It was found that the dependence of the ratio of HG00 light to the total light depends mainly on the focal length of SR3 and is not effected by the presence of arm cavities. This is because the arm cavities are well matched to the input

laser [82]. When the arm cavities are on resonance, the ITMs act as highly reflective mirrors with frequency dependent behaviour. This means that there is no difference in HOM content of the beam transmitted through the SRM between a single bounce measurement and an arm cavity measurement.

The HOM content of the light does depend on the detuning of the SRM. It does not depend on whether the interferometer is in signal recycling mode or RSE mode (as the arm cavities do not matter). When the SRC is on resonance, it is much more sensitive to an error in the SR3's focal length than when the SRC is anti-resonant. The light is less sensitive to an error in the focal length of SR2.

The beam parameter of the arm mode at the SRM behaves in a predictable way since the arm mode is well defined, so the Gouy phase of a beam at the SRM can be used to acquire information about the arm mode at the SRM. The beam width and RoC were simulated for a range of SR3 focal length errors. By analysing where the SRC becomes unstable, the beam size cannot exceed 2.2 mm. These results can be used to constrain design tasks related to active wavefront control. The Gouy phase has been measured to be $21.3 \pm 0.3^\circ$ [83]. If the thermal lensing in the ITM is roughly at design, then the focal length of SR2 is -3.205 ± 0.003 m and the focal length of SR3 is 17.971 ± 0.002 m. The arm mode would be 1.85 mm and a curvature of -3.8 m and there would be a few percent HOM in OMC scans. **Double check values, need extracting from the plot.**

2.1.1 Measurements of the Mode Matching Between the Arm Cavities and the OMC at LLO Between 2018-2019.

The mode matching between the arm Cavities and the OMC, and the input light to the OMC has been studied at LLO. The following subsection describes measurements made at LLO between 2018-2019. Techniques for determining the mode matching between the arms and the OMC include: actuating the waist size and waist location of the beam coming out of the arm and measuring what effect this has on the mode matching to the OMC; direct beam profile measurements; the application of different DARM offsets to determine what portion of light in an OMC scan corresponds to the arm mode and what portion of the light comes from elsewhere in the interferometer (e.g. from contrast defect); and use of a beacon signal to distinguish light from the signal-carrying arm cavity light from junk light. Modelling work relies on the modematching of the PRC and the arms to be $> 99\%$ (estimated from interferometer reflectivity [82]), as this allows singlebounce and full interferometer measurements to be compared. The upper limit on mode matching from the arm to the OMC is 99% , and this was obtained from actuation on the beam parameter and the use of a beacon signal (meaning this must be arm cavity light). The entire collection of measurements find the mode matching to be between 87% and 99% . Since this is quite a large discrepancy, the measurements are briefly discussed below. *Analysis of these results is being developed (cite Carl Blairs work)?.*

Mode scans for an input power of 10W find the mismatch to be 11% [84]. The DARM offset was changed in order to determine what light was coming from the arm mode. An input power of 10W is fairly low, the design is 125W , so the thermal

state of the interferometer may be different in this measurement compared to when the interferometer is in observing mode. This method was repeated for 40W of input power [85]. In this measurement, it was found that there was 4% mode mismatch.

A direct measurement of a beam profile finds there to be A 15% mode mismatch between the OMC waist and the beam transmitted through the SRM while the interferometer was in an ITM single bounce [86] [87] configuration. This measurement also shows a fair amount of astigmatism, with the waist in the x and y directions being separated by ≈ 0.5 m. The x direction data is discarded in the analysis as there was trouble orienting the beam profiler. Since there is good mode matching between the input and the arms ($> 99\%$), there should not be much difference between this beam and the beam when the interferometer is locked as long as the thermal lensing in the ITM was similar to how it would be in observation mode. This type of measurement cannot be performed while the arms are locked as they require commissioner to be working nearby the arm cavities. This analysis is in fairly close agreement with the mode scan measured in [84].

Altering the beam parameter using the ITM thermal lens actuator has shown that the mode matching between the arm cavity and the OMC can be improved by 1% or 2% [88]. Using the SRM CO2 heater increased the optical gain [88], and the amplitude of the beacon signal increased by 10%. The ITM and the SRM are separated by enough Gouy phase that actuating on them covers a sufficient amount of beam parameter phase space in determine if the mode matching was optimised. does [88] agree with [84] and [86] [87]. The interferometer was in single bounce mode, but since the mode matching between the PRC and arm Cavity is high, this

is similar to a locked-arm cavity measurement. There was 25W of input power, so thermal effects may be different between this measurement and observation mode. This type of measurement was repeated using 40W of input power [89]. Assuming the ITM thermal lens was optimal, then the mode matching from the arm to the OMC was found to be optimal with no SRM heating applied. This suggests that the mode matching is $> 99\%$. A beacon signal was used to determine the mode matching from the arm to the OMC.

Modelling of the arm mode finds that a 3.5% mode matching improvement between the arm cavity and the OMC can be made using the ITM thermal lens and the SRM thermal lens [90]. This modelling is compared to the measurement made with data from [87]. It was found experimentally that a 4% improvement in the arm to OMC mode matching could be achieved for a single bounce measurement with 25W of input power. This evidence is in disagreement with [89] which shows that the mode matching is already high.

Just after the interferometer had lost lock, a series of mode scans were used to create a model which predicts 92% mode matching [91] between the beam transmitted through the SRM and the OMC. The input power was 20W and the lock lasted 1 hour and 41 minutes, so the interferometers thermal state immediately after this lock loss was the same as it would be for a typical time during observing mode when the input power was 20W. The thermal lens in the ITM was measured using the Hartmann wavefront sensors, and this was used to model the interferometer at the time of unlocking. The Hartmann data is very noisy, which could cause uncertainty in the mode matching prediction made by the model, although no error on the fit

parameters is given in [91].

The mode matching between the squeezer and the optics within HAM6 was studied [92] [93]. Measuring the beam profile between the squeezer and the OMC allows for the layout in HAM6 to be modelled. This means that there is less uncertainty in where the mismatch between the arms and the OMC may come from. The mode matching between the squeezer and the OMC was measured to be 98%. The model of HAM6 that was developed from this data differs significantly from the nominal values of the optics in HAM6, a 15% error in the RoC of OM2 is required to explain the beam profile measurements. This may not be unreasonable, as accurate measurements of focal lengths for mirrors with $f = 2\text{m}$ are hard to make (ref?).

2.2 Modelling Interferometers With Mode Mismatches

Using FINESSE

Mode decomposition and mode mismatching will be described earlier in thesis. This section is specific to FINESSE.

FINESSE is software for simulating optical layouts. Optical layouts are modelled with components such as mirrors, lenses, and spaces. The way these components are connected is defined by nodes. FINESSE performs a nodal analysis to calculate the steady state amplitude of light fields within the optical layout, similar to how a circuit can be analysed. PyKat is a python wrapper for FINESSE which provides many quality of life improvements such as the ability to programatically updating and re-running simulations [94].

The light fields that are calculated in a FINESSE simulation can be measured by using detectors such as a photodiode, `pd`, or an amplitude detector, `ad`. These can be configured to detect only certain modes using the `mask` command. The `bp` detector is used for measuring the beam parameter, and the `gouy` command computes the Gouy phase accumulated along a set of spaces. These detectors were used to find the ratio of HOM to HG00 light, the beam parameter of the arm mode that is transmitted through the SRM, and the Gouy phase accumulated along the SRC.

The beam parameter at nodes in a FINESSE simulation are determined by `cav` and `gauss` commands. The `cav` command is used for defining linear and circular cavities. The beam parameter is found by solving for the cavity eigenmode. Once these nodes have had a beam parameter defined at them, the components connected to them have their beam parameter calculated based on the set of linear equations that define the component. If there are more than one `cav` commands used in a simulation, the latter `cav` command overwrites the previous one. The `gauss` command executes after the `cav` commands, and sets the beam parameter of the laser used in the simulation.

FINESSE can perform simulations using HG modes, allowing for the effects of mode mismatches to be modelled as long as the problem satisfies the paraxial approximation. A coupling between the HG modes happens whenever a beam described by q_1 enters a segment described by q_2 . This could happen if the `gauss` command used on the laser does not correspond to the eigenmode of a cavity defined by the `cav` command.

A laser beam can be modelled as a linear combination of Hermite Gaussian modes. Ideally an infinite number of these modes would be used in a calculation, however since it is not feasible to simulate an infinite number of TEM modes the maximum number of TEM modes needs to be defined in a FINESSE simulation. This is done using the `maxtem` command. FINESSE will calculate the coupling of the light amongst these modes. If an insufficient number of TEM modes is used in the simulation, the output will be nonsense as power from the higher order TEM modes that would be required to accurately model the optical layout will be ‘aliased’ into the ones defined in the simulation. When a cavity becomes unstable, FINESSE will not generate physical results since the cavity will not be mathematically describable.

The distance between components, L , in a FINESSE simulation is defined with a macroscopic distance, l , which is modelled with the `space` command, and a microscopic distance which is modelled with the tuning parameter, ϕ , belonging to components such as mirrors (`m`) and beamsplitters (`bs`). The macroscopic distance l is always an integer number of wavelengths, whereas the microscopic tuning, ϕ , determines the sub-wavelength accuracy of the length of the space. FINESSE uses the convention that 1 wavelength corresponds to 4π rad phase. It is useful to split the total length, $L = l + \phi$, in this way since the l determines things like Gouy phase, whereas ϕ determines whether a cavity is on resonance/which mode is resonant. When simulating mode mismatches, the command `phase 2` is required to give physically correct results. This sets the phase accumulated by the HG00 mode in a `space` to 0° .

2.2.1 Thin and Thick Beamsplitters

Thin beamsplitters are made with the `bs` command. Care needs to be taken when using these components since incorrect usage can yield erroneous results. To illustrate this two simulations of Michelson interferometers were run, one with a thin beam splitter and the other with a thick beamsplitter. The FINESSE code for these simulations can be found in appendix ??.

For energy to be conserved, each reflection from a dielectric coating where the refractive index is lower on the reflecting side than on the transmitting side, a light field gains 180° of phase. In a Michelson, there is one path with 2 of the π phase change reflections, while in the other path there is only one such reflection. This means when the arms have equal lengths, destructive interference occurs at the AS port. The intensity, I , at the AS port of the Michelson as a function of the tuning of one of the arm mirrors, $\Delta\phi$, including the FINESSE convention for phase, is

$$I = I_0 \sin^2(\Delta\phi) \tag{2.1}$$

where I_0 is the input power.

The effect of using a thin beamsplitter versus a thick beamsplitter in a FINESSE simulation is shown in figure 2.1. The thin beamsplitter causes there to be constructive interference at the AS port for equal arm lengths, whereas equation 2.1 shows that it should be destructive interference. Using a thick beamsplitter fixes this ‘bug’, as the phase at the beam splitter is properly accounted for.

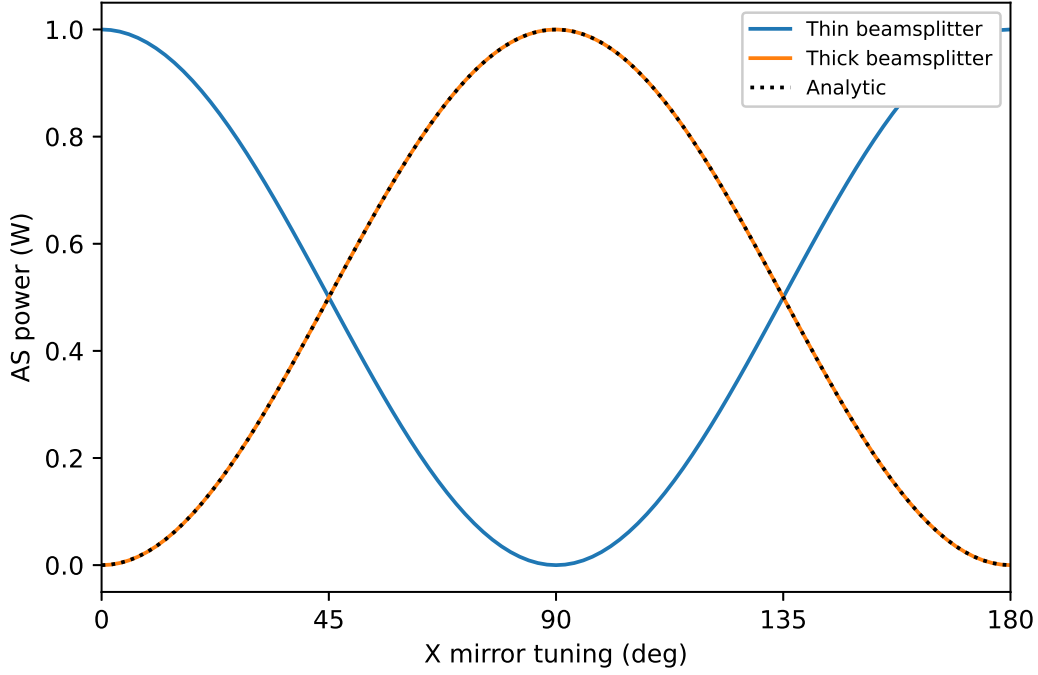


Figure 2.1: The simulation of a Michelson interferometer with thin a beamsplitter is shown in blue. The same simulation was done with a thick beam splitter, this is shown in orange. The analytic behaviour of a Michelson interferometer (equation 2.1) is shown by dashed black line. The tuning of one of the arm mirrors was changed and the power at the AS port was measured. Comparing the outputs of the simulation to equation 2.1 shows that thick beamsplitters should be used.

2.3 Modelling of Michelson Interferometer With Fabry-Perot Arm Cavities and A Signal Recycling Cavity

To investigate what effect a mode-mismatch between the SRC and the arm cavities has on the HOM content of the beam transmitted through the SRM, a ‘LIGO-like’ interferometer was simulated in FINESSE. This interferometer uses identical, design specification arm cavities and the LLO SRC. The reason for using identical, design value arm cavities is to eliminate contrast defect. The OMC, IMC, and PRC were not included in this simulation as these do not effect the beam in the SRC. To perform ITM single bounce measurements, the ETMs are misaligned resulting in the ITM having an effective power reflectivity of 1. To replicate this in FINESSE, the R for each ITM was set to 1 minus the loss of the ITM HR coating. This is valid as long as the mode-matching between the PRC and the arms is close to 100%. The interferometer was set up in DC readout mode. The tuning of the arm mirrors was set so that there was 30 mW of light emerging from the AS port with nominal spacings and optic RoC. A schematic of the simulation is shown in figure 2.2. Nominal parameters in the simulation are listed in table ??.

Geometry errors in the SRC cause couplings between the HG00 mode and higher order modes. There are two sorts of these errors: RoC errors and spacing errors. RoC errors can be found in SR3, SR2, SRM and the off-design thermal lensing in the ITM and compensation plate, and spacing errors between each of the optics in the SRC. The largest geometry errors in the SRC come from RoC errors in SR3 and

SR2, with both errors affecting the beam in a similar way as there is not much Gouy phase between them. The thermal lensing in the ITM is assumed to be 50 km, and it would require significant deviation, in the direction of more heating, for this lens to have a large effect. The radius of curvature of SRM is well enough known that it is not the dominant cause of junk light in the SRC. The spacings between the optics in the SRC is known to a great enough precision that errors in their locations have a negligible effect. Therefore, to investigate how geometry errors effect the beam transmitted through the SRM, the RoC of SR3 and SR2 will be varied.

All simulations used the same spacings, RoC and focal lengths, apart from the focal lengths of SR3 and SR2 which was a variable for most simulations. The ETM and ITM both have 10 ppm loss. The arms are both made to be identical so there is no contrast defect, and are based off the nominal design. A 50km thermal lens is assumed in all simulations. The lengths in the SRC are ones directly obtained from the Zemax model of the LLO interferometer. To simplify the modelling, the SRC telescope is constructed using `lens` rather than `bs` with `attr` commands (although its not much more complex to use `bs`). The focal lengths in the SRC are the measured values for the LLO optics.

For simplicity, no Schnupp asymmetry was included. In a real interferometer, a Schnupp asymmetry is required to make the interferometer controllable. Since spaces are by default on resonance in FINESSE, no Schnupp asymmetry is needed to control the inner michelson length. The Rayleigh legnth of the beam at the ITM is of the order of several hundred metres which is huge in comparison to the Schnupp asymmetry, which is 8 cm. The Schnupp asymmetry will have a small effect on the

HOM content of the beam transmitted through the SRM, but since the dominant source of HOM comes from RoC errors in SR3 and SR2, the behaviour of the beam in the SRC can be largely understood without it.

The beam parameter of the input laser was defined so that there is no difference between the single bounce and the locked arm simulation. To define the beam parameter, the **gauss** command was used. In this simulation, the mode matching from the laser to the arm cavities should be perfect, the actual amount used in this simulation is 99.99999%. See appendix ?? for details on how w_0 and z were determined for the **gauss** command used on the laser. The power of the input laser was set to 5 MW.

The **cav** command was used to set up the arm cavities, although the **gauss** command used on the laser means the **cav** command is not required (on the fly testing showed this to be the case). A **cav** command was not used on the SRC since this forces the beam parameter at the SRM to be determined by the SRC geometry. An SRC cannot be defined in FINESSE using the **cav** command as one of the end mirrors cannot be properly defined.

The detectors used to analyse the mode-matching between the nominal arm mode (which matches the laser mode) and the SRC mode were **pd** and **ad** with various **mask** commands applied to them. To detect the total amount of light, no **mask** is applied. To detect the amount of higher order light a **mask [pd] 0 0 0** command is used. This sets the sensitivity of the photodiode to the HG00 mode to 0. The mode matching is then determined as $1 - \text{HOM}/\text{total light}$. **maxtem** is set to 6. As a cross check, 2 amplitude detectors were used to detect the HG02 and HG20 modes:

```
ad ad_HOM_2_y 2 0 0 nSRMb
ad ad_HOM_2_x 0 2 0 nSRMb
```

When plotting the power of the HG20+HG02 mode, the square of the `ad` detector output needs to be taken. The `bp` detector was used to measure the beam parameter. To measure the gouy phase accumulated in the SRC by the HG00 mode, the `gouy` command was used.

Both the signal recycling and RSE modes of operation were investigated. The tuning of the SRM determines whether the interferometer is in RSE mode or in signal recycling mode. If the interferometer has arm cavities, the SRM detuning which corresponds to RSE is 90° while the detuning which gives signal recycling is 0° . For an interferometer without arm cavities, to operate in RSE mode the SRM detuning should be set to 0° , while to be in signal recycling mode the detuning of the SRM should be set to 90° .

2.3.1 Determining Tunings for DC readout

To maintain the same SRC finesse seen by the differential mode for each simulation, the interferometer set up so that it was on the dark fringe with 30 mW of light being transmitted through the SRM. This was achieved by applying a differential tuning to the ETMs for simulations with the arm cavities, and a differential tuning to the ITM for single bounce simulations. The light transmitted through the SRM for an interferometer with arm cavities as a function of differential ETM detuning is shown in figure 2.3. For an interferometer without arm cavities, the power transmitted through the SRM as a function of differential ITM detuning is shown in figure 2.4.

The tunings used to operate with 30 mW of light being transmitted through the SRM are shown in tables 2.1 and 2.2.

Mirror	SRM Tuning ($^{\circ}$)	Mirror Tuning ($^{\circ}$)
ETM _x	0	$+2.76\text{e} - 06$
ETM _y	0	$-2.76\text{e} - 06$
ETM _x	90	$+2.64\text{e} - 05$
ETM _y	90	$-2.64\text{e} - 05$

Table 2.1: The tunings used for the arm-cavity simulation to get 30mW of light at the dark port.

Mirror	SRM Tuning ($^{\circ}$)	Mirror Tuning ($^{\circ}$)
ITM _x	0	$+0.00709$
ITM _y	0	-0.00709
ITM _x	90	$+0.00074$
ITM _y	90	-0.00074

Table 2.2: The tunings used for the no-arm-cavity simulation to get 30mW of light at the dark port.

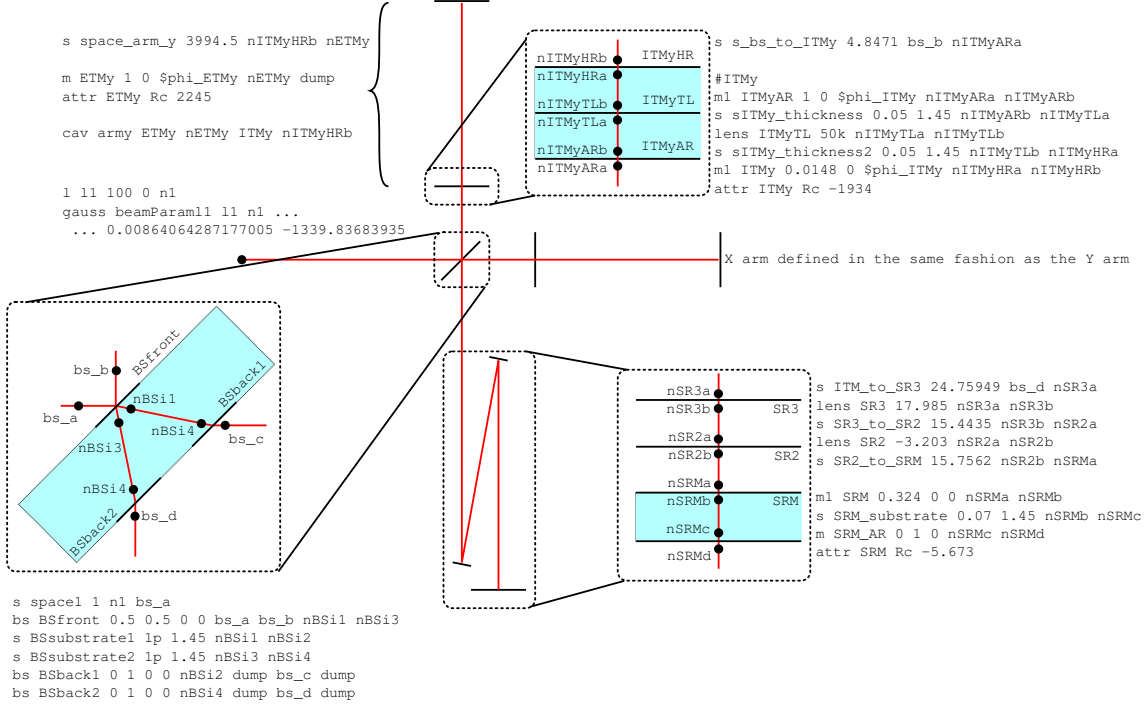


Figure 2.2: Schematic of the simulation. The dimensions of the schematic are not to scale. The laser light is shown as a red line. Black lines with red beams interacting with them corresponds to components such as mirrors and lenses. Important nodes are highlighted with black dots and the associated FINESSE commands are shown nearby. Zoom-ins provide more detail for the main beam splitter, ITMy and the SRC. The elongated curled bracket highlights the arm cavity, and the text with it shows how it is implemented in FINESSE.

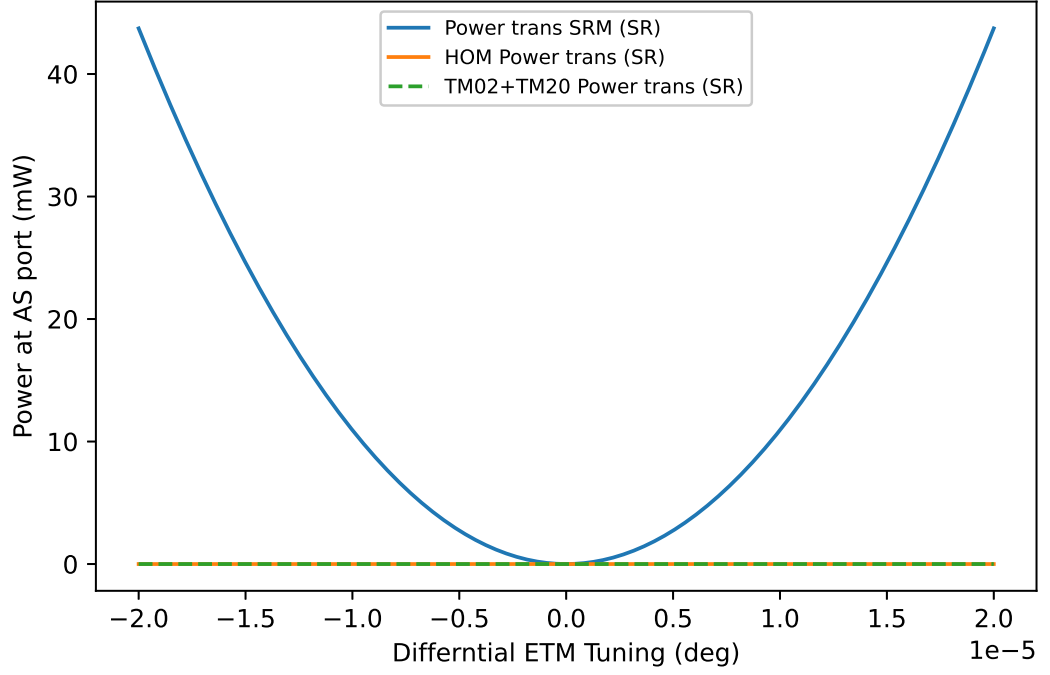


Figure 2.3: The minimum of the dark fringe is at 0° differential ETM detuning, as shown by the blue line. A slight offset causes light to be as the AS port. This result was used to determine tunings for simulating what effect focal length errors in SR3 and SR2 have on the beam transmitted through the SRM. The orange and green dashed line show the HOM content of the beam.

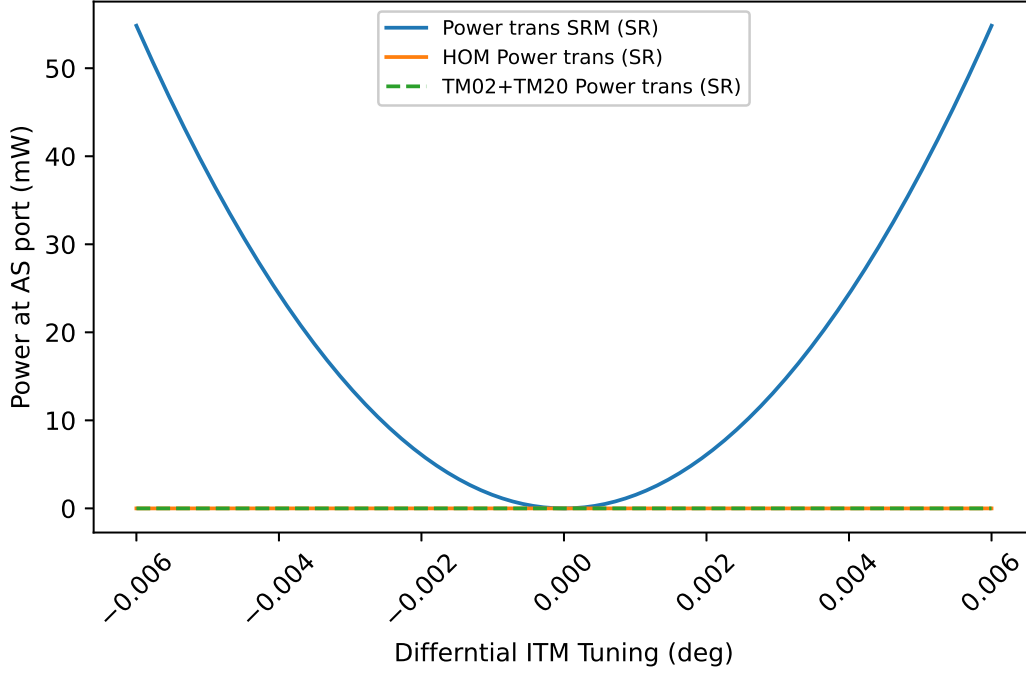


Figure 2.4: The minimum of the dark fringe is at 0° differential ITM detuning, as shown by the blue line. A slight offset causes light to be as the AS port. This result was used to determine tunings for simulating what effect focal length errors in SR3 and SR2 have on the beam transmitted through the SRM. The orange and green dashed line show the HOM content of the beam.

2.4 The Beam Parameter of the Arm Mode at the SRM

The mode defined by the arm was measured at the SRM. The beam parameter at the SRM depends on macroscopic quantities such as the distances between mirrors and the RoC of the optics, rather than microscopic tunings. Therefore, the beam parameter is not affected by the interferometer being in RSE or signal recycling mode. The beam parameter depends only on the geometry of the cavities. The focal length of SR3 and SR2 were varied around 3 standard deviations of the measured values, and the effect this had on the beam transmitted through the SRM was measured with a **bp** detector.

The effect of a focal length error in SR3 and SR2 on the width and RoC of the arm mode at the SRM is shown in figure 2.5, and 2.6. These figures show that the beam parameter depends mainly on the focal length of SR3, although SR2 has a noticeable effect. When the focal lengths of SR3 and SR2 is at the nominal values, the simulated beam has a width of 2.1 mm, a RoC of -4.0 m and the Gouy phase accumulated between the ITM and the SRM is 18.7° . There is a slight difference between the RoC found here and the design value of -3.88 m, this is because the focal lengths of the LLO SRC optics is not the same as the design values. Measurements of the single bounce beam profile [87] suggests that the beam is much larger at the SRM than the design value. This measurement also found the beam to be astigmatic, with $w_y = 2.6$ mm and $w_x = 2.4$ mm. This would correspond to a positive error in the focal lengths of SR3, as shown in figure 2.5. If the beam was this large it is unlikely that the SRC would be stable, as shown in figure 2.9.

The Gouy phase as a function of focal length error in SR3 and SR2 is shown in figure 2.7. Measurements of Gouy phase of the SRC find it to be 22.81° [95] and 21.3° [83]. These measurements of Gouy phase suggests that the beam at the SRM is smaller than the design value, and has a lower curvature. This amount of Gouy phase also corresponds to the same order of magnitude of HG00 loss that is measured and simulated (figure 2.12). The Gouy phase accumulated in the SRC is almost entirely dependent on the RoC of SR3 and SR2, positioning errors of up to 1 cm of SR3 and SR2 do not effect the Gouy phase significantly. Figure 2.7 shows that it is highly unlikely that the Gouy phase could be 22.81° in a state where the arm has greater than 90% mode matching to the SRC, as the RoC of both SR3 and SR2 would have needed to have been three standard deviations from the measured value. This could be accounted for by an much shorter thermal lens in the ITM (see figure 2.8). The measurement of 21.3° of Gouy phase is plausible with the 50 km thermal lens. The single pass thermal lens was measured to be 38.92 uD or ≈ 25.7 km [91], which corresponds to the design value and the value used in this simulation.

The focal length of SR3 and SR2 can be estimated from the Gouy phase measurement. A normal probability density function was used to model the uncertainty in the focal length of SR3 and SR2. This was combined into a 2-D normal distribution. Combinations of SR3 and SR2 that result in 21.3° of Gouy phase were found and the value of the 2-D normal distribution for those combinations was calculated. The maximum value of the resulting probability density function (which is normal) corresponds to the most likely combination of SR3 and SR2, and one standard deviation corresponds to the error. Figure 2.10 shows the that focal length of SR2 to

be -3.205 ± 0.003 m and the focal length of SR3 to be 17.971 ± 0.002 m.

The beam parameter at the SRM is determined by the focal length of SR3 and SR2. Each beam width and RoC that was computed in this simulation has a probability density associated with it (as this depends on SR3 and SR2, see figure 2.10). Therefore, the probability density function for the beam parameter of the arm mode at the SRM can be computed. Based on the measurement of Gouy phase, the beam is likely to have width 1.84 mm and a defocus of -2.80 Dioptre . This is shown in figure 2.11. This is a slight simplification as the beam parameter cannot be such that the Gouy phase of the cavity becomes too small ($< \sim 17.5^\circ$). IF the Gouy phase is too small the cavity will be unstable, so beam widths of more than 2.25 mm are very unlikely. The shape of the distribution of the arm mode beam parameter at the SRM is driven mostly by the focal length error of SR3.

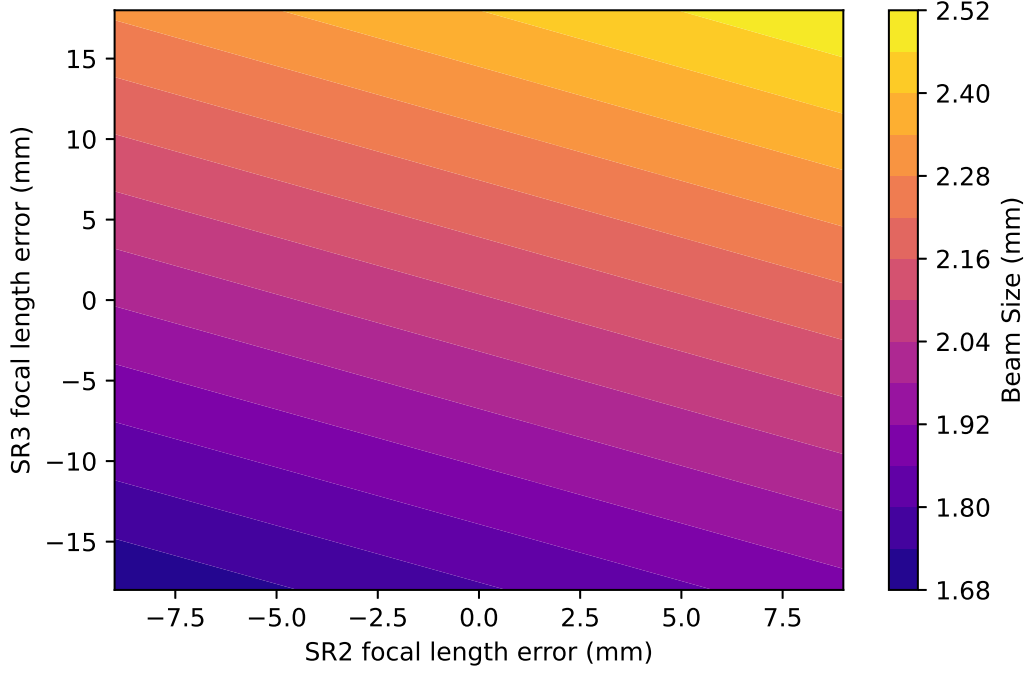


Figure 2.5: The Gaussian width of the beam that is transmitted through the SRM as a function of the error ($\pm 3\sigma$) in focal length of SR3 and SR2. The Gaussian beam width is strongly dependent on the error in SR3’s focal length, whereas SR2’s focal length error is less important. When the SR3 focal length error > 6 mm and SR2 error is zero the SRC cavity is unstable (see figure 2.9), so this area of the parameter space is unlikely to correspond to the beam at the SRM at LLO.

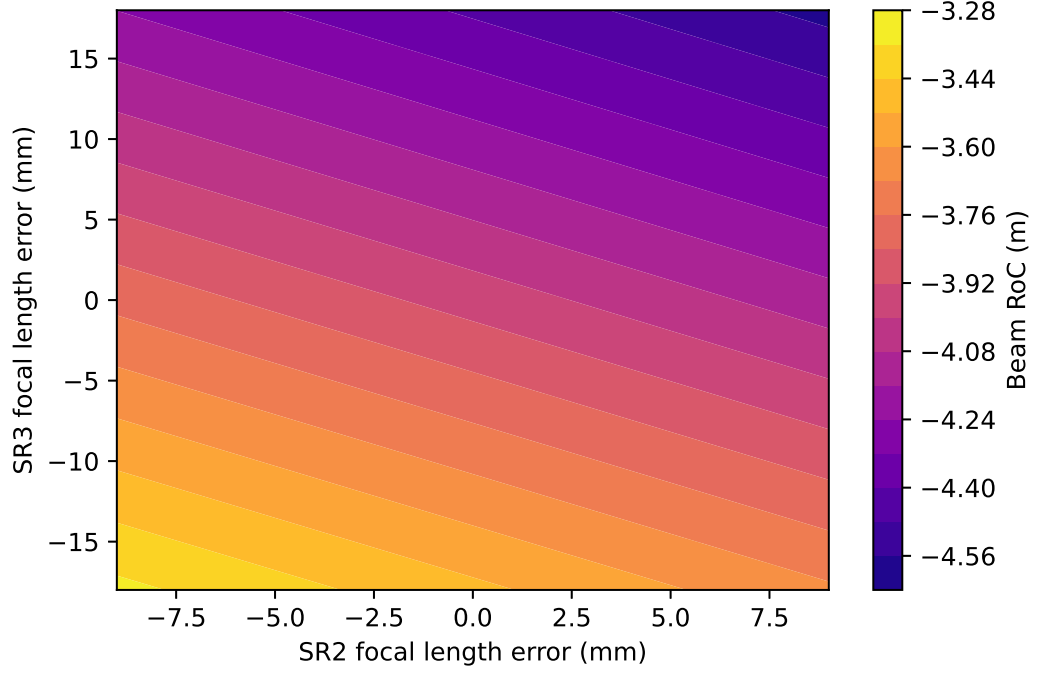


Figure 2.6: The RoC of the beam that is transmitted through the SRM as a function of the error ($\pm 3\sigma$) in focal length of SR3 and SR2. If the arm cavities and the SRC are well mode matched, then the RoC of the beam transmitted through the SRM would be $\frac{5.67\text{m}}{1.45} = 3.91\text{m}$. When the SR3 focal length error > 6 mm and SR2 error is zero the SRC cavity is unstable (see figure 2.9), so this area of the parameter space is unlikely to correspond to the beam at the SRM at LLO.

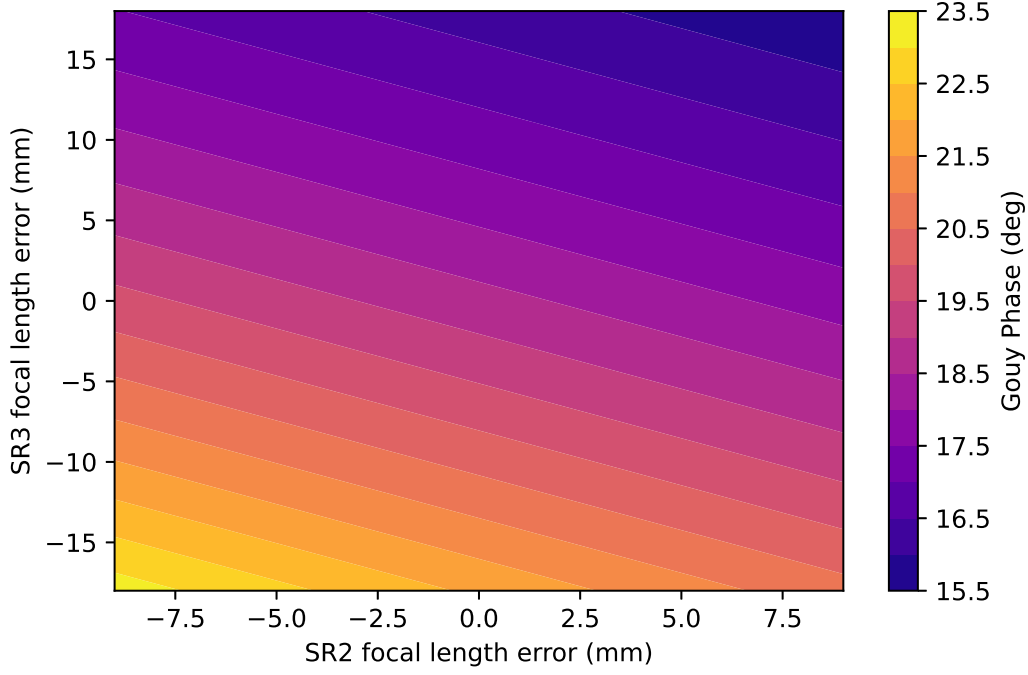


Figure 2.7: Gouy phase between the ITMs and the SRM as a function of the error ($\pm 3\sigma$) in SR3 and SR2. When the SR3 focal length error > 6 mm and SR2 error is zero the SRC cavity is unstable (see figure 2.9), so this area of the parameter space is unlikely to correspond to the beam at the SRM at LLO.

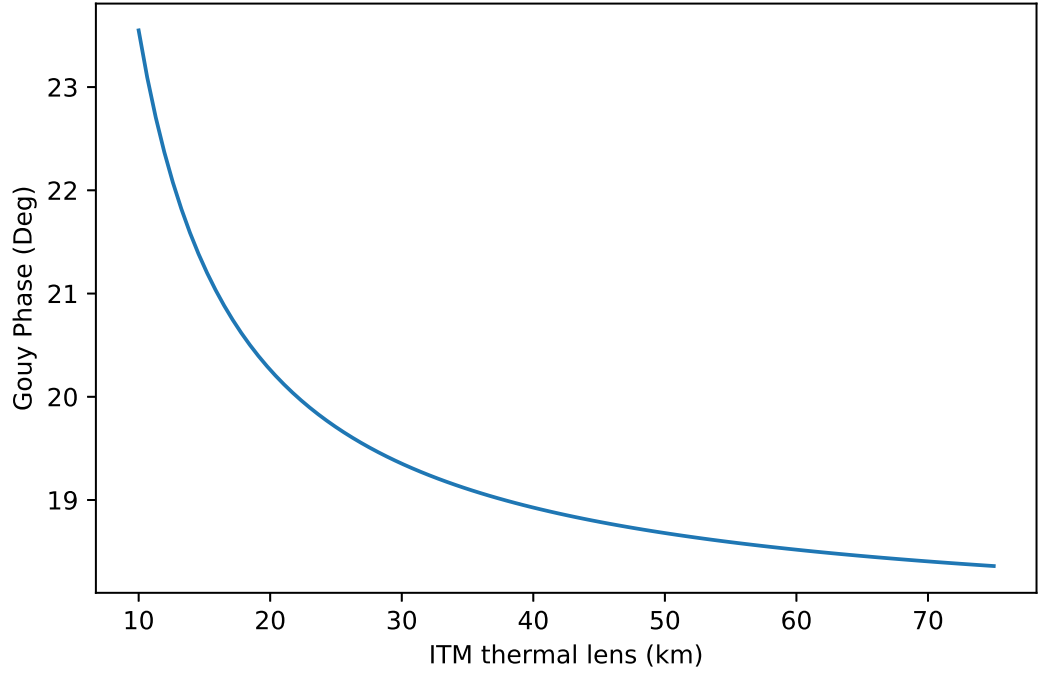


Figure 2.8: Gouy phase between the ITMs and the SRM as a function of the thermal lens in the ITMs for nominal values of SR3 and SR2. The nominal ITM thermal lens is 50 km, and this corresponds to about 19°. More heating results in a greater thermal lens, however at LIGO this is compensated for by the thermal compensation system (TCS).

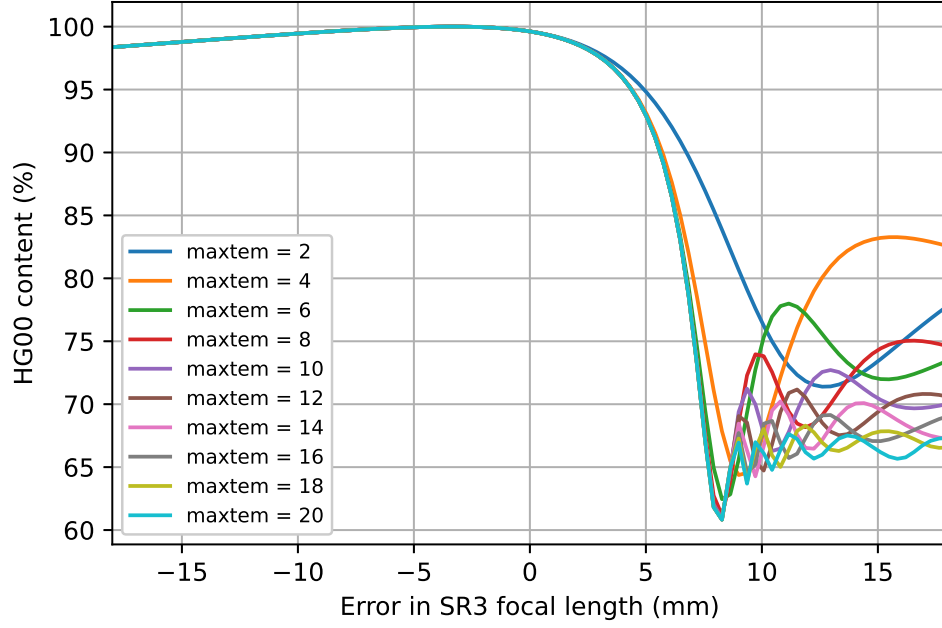


Figure 2.9: The percentage of HG00 light transmitted through the SRM as a function of SR3 focal length for the Michelson interferometer with Fabry-Perot cavities in its arms was simulated. Each simulation used a different `maxtem` command. The focal length of SR3 was changed. For focal lengths with a positive error greater than 7.5 mm, the cavity becomes unstable and FINESSE does not produce reliable results. For a stable SRC, the minimum number of TEM modes for sub 1% accuracy is six.

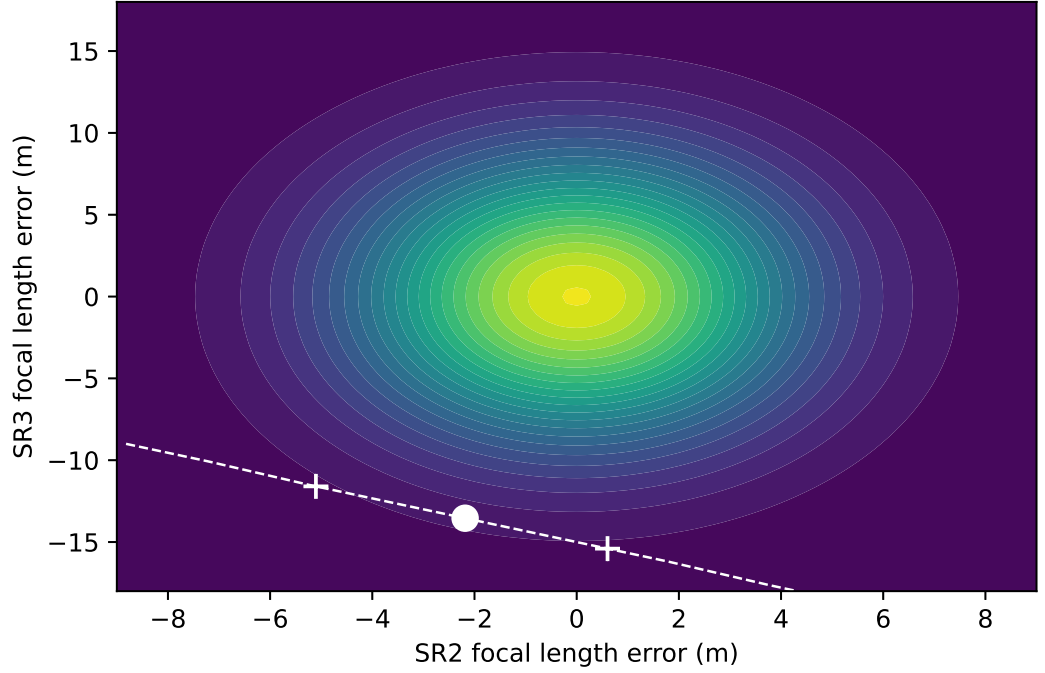


Figure 2.10: The measurement error of SR3 and SR2 follows a normal distribution. These were combined to make a 2-D normal distribution. Dark blue corresponds to the least likely combinations of SR3 and SR2, and bright yellow corresponds to the most likely. The measured Gouy phase was 21.3° [83], this is indicated by the white dashed line. The white circle marks the most likely combination of SR3 and SR2 which gives 21.3° of Gouy phase, and the white crosses show 1 standard deviation from this. The white circle and crosses indicate that the focal length of SR2 is -3.205 ± 0.003 m and the focal length of SR3 is 17.971 ± 0.002 m.

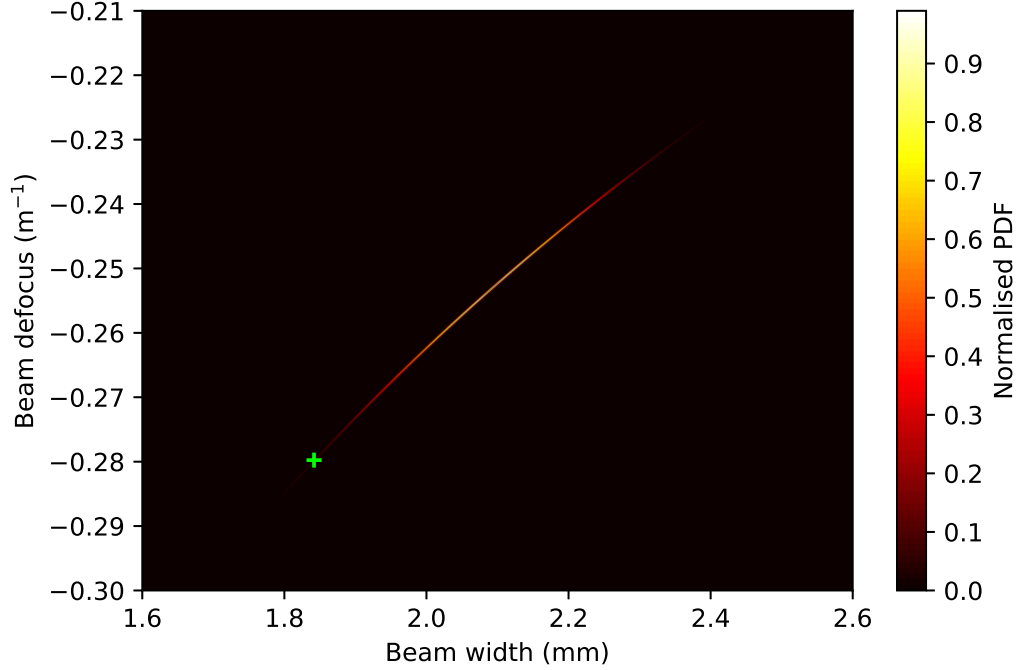


Figure 2.11: The beam parameter at the SRM was calculated for combinations of SR2 and SR3 focal lengths. The associated probability density for these combinations can be calculated (see figure 2.10), so the probability of the beam parameter having a certain width and defocus can be calculated. The probability density for each pair of beam width and defocus is indicated with the colour bar. The measured Gouy phase of 21.3° [83] is indicated with a green cross (1.84 mm, -2.80 Dioptre). This is a slightly simplified approach since the beam parameter cannot be such that the SRC is unstable. This means certain combinations of beam width and beam defocus are not possible. Beam parameters with a width $> \sim 2.25$ mm result in an unstable SRC, therefore these beams are very unlikely to be present at the SRM at LLO.

2.5 Power Transmitted Through SRM for a Mode Mismatch Between the Arm Cavity and the SRC

The HOM content of the beam as a function of the error in focal length in SR3 was investigated. As the arm cavities become more mismatched to the SRC, less HG00 mode is transmitted. The mode mismatch can be seen in figure 2.6. As the focal length of SR3 becomes further from its nominal value, the radius of curvature of the beam at the SRM stops matching the radius of curvature of the SRM. Since the SRC is a low finesse cavity, a significant amount of mismatch can be tolerated before there is a significant drop in the HG00 power transmitted through the SRM. This is shown in figure 2.12. Figures 2.12 and 2.13 show that the nominal value of the focal length SR3, 17.985 m, is slightly longer than optimal. When the SRC is on resonance, the peak in transmission is -3 mm from the nominal focal length of SR3. This is shown by the blue and green lines in figure 2.13.

The power of the HG00 light transmitted through the SRM as a function focal length error is shown in figure 2.13. In signal recycling mode, shown as blue and green in figure 2.13, the transmitted power has a peak shape. This is because an error in the focal length of SR3 moves the cavity off resonance. In RSE mode, shown as red and orange in figure 2.13, a change in focal length of SR3 does not have much effect on the amount of HG00 transmitted by the SRM since an anti-resonance is a much broader feature than a resonance.

In a single bounce measurement, the beam incident onto the ITM from outside the cavity is reflected straight back. This is equivalent to the reflectivity of the ITM

$R = 1$. When the light is resonant in the arm cavity, the ITM acts as a highly reflective mirror with some frequency dependent behaviour. This means that there is no significant difference between a single bounce measurement and a measurement with arm cavities. This is shown in figure 2.12. Higher order mode beams generated by errors in the SRC are not suppressed by the arm cavity as they are reflected by the ITM.

The ratio of the amount of light in the HG00 mode versus the total light depends on the detuning of the SRM, rather than the presence of cavities in the arm. This is shown in figure 2.12. When the light enters the arm cavities, it is reflected back with an extra π rad phase, however the HOM that does not enter the cavity does not acquire this phase. Since the HOM only interact with the SRC, the ratio of HOM to HG00 depends on the detuning of the SRM, rather than whether the interferometer is in signal recycling or RSE mode. The amount of higher order mode generated in the SRC is proportional to the HG00 power in the SRC, so the percentage of HOM beam is dependent only on the tuning of the SRM and the focal length error of SR3.

As the SRC gets closer to being unstable, more TEM modes are required to simulate the behaviour of the cavity. The effect of using more TEM modes in the simulation is shown in figure 2.9. For positive errors above 6 mm in the focal length of SR3, the cavity becomes unstable. As the error in SR3 focal length approaches this, a greater number of TEM modes is required to accurately simulate the behaviour of the interferometer. It was found empirically that six TEM modes is sufficient to simulate the interferometer while it is stable. The greyed out regions of figures 2.12 and 2.13 show where the simulation is not accurate.

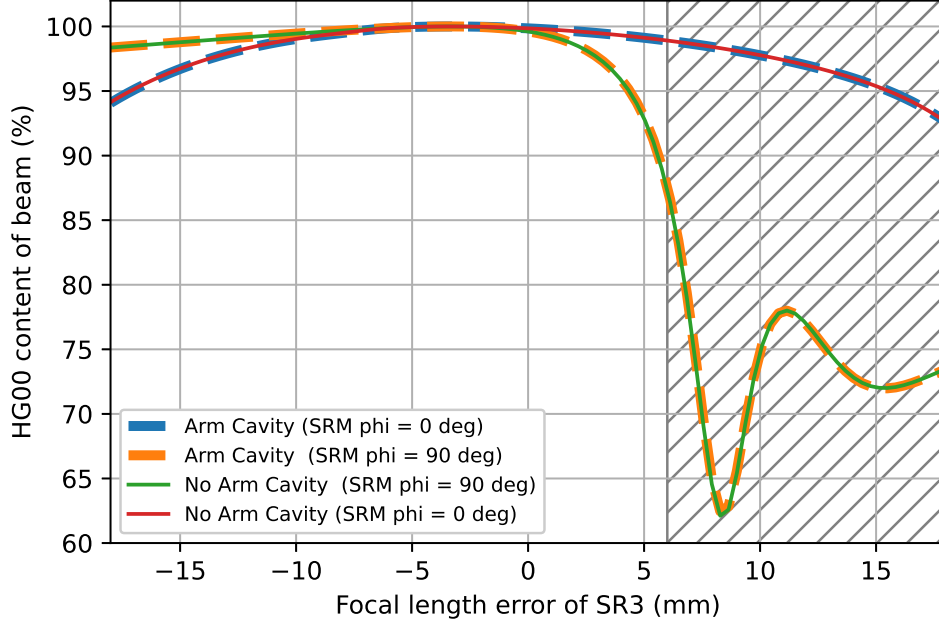


Figure 2.12: The percentage of light which is in the HG00 mode as the focal length of SR3 changes compared to the total light power does not depend on whether there are arm cavities. It is determined by the detuning of the SRM. The blue dashed line shows the behaviour of the beam with arm cavities present and with the SRC in signal recycling mode (SRM $\phi = 0^\circ$). The red line shows how the beam changes with focal length error for a single bounce measurement and with the SRC in RSE mode (SRM $\phi = 90^\circ$). The orange dashed line corresponds to the interferometer having arm cavities and being in RSE mode (SRM $\phi = 90^\circ$). The green line shows behaviour of the beam when the interferometer is configured for a single bounce measurement and is in signal recycling mode. The hatched region indicates where the SRC becomes unstable (see figure 2.9), so the behaviour of the interferometer cannot be accurately simulated for these values of focal length.

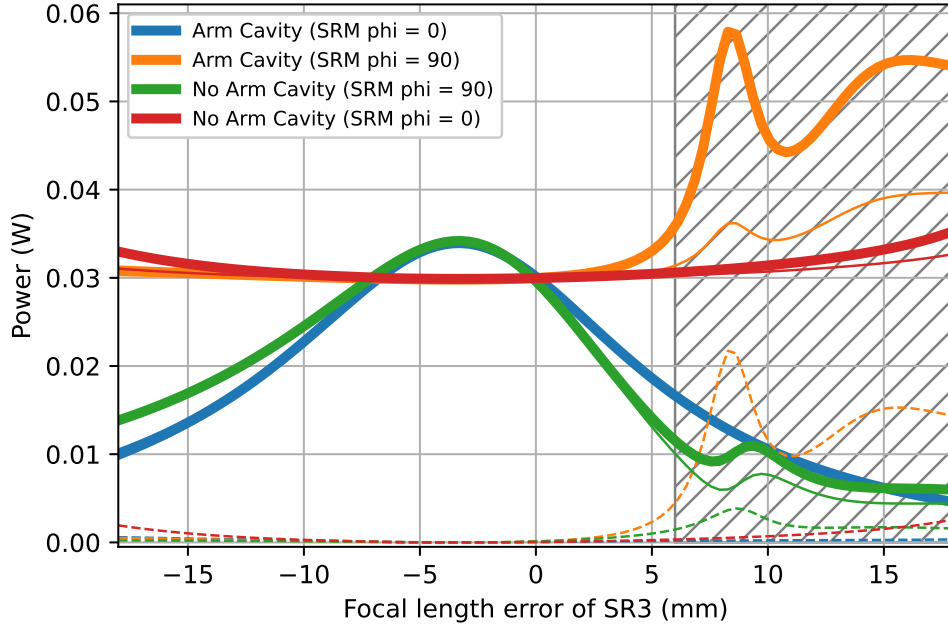


Figure 2.13: The total power transmitted through the SRM for each simulation, is shown in the thick pale lines. The thin solid lines corresponds to the power in the HG00 mode, and the thin dashed lines correspond to the power in the higher order modes. Blue lines show the behaviour when the interferometer has arm cavities and is in signal recycling mode, and the orange lines shows when it is in RSE mode. Green lines show how the power changes when the interferometer is configured for a single bounce measurement and is in signal recycling mode, and the red line shows when its in RSE mode. The hatched region indicates where the SRC is becomes unstable, and so the behaviour of the interferometer cannot be accurately simulated for these values of focal length.

2.5.1 Change of finesse of SRC

As the transmission of the SRM decreases, the finesse of the cavity increases. A FINESSE simulation was run to see the effect of increasing the power transmission of the SRM. The HG00 mode power decreases with lower SRM transmission slightly slower than the power transmitted in the HG20+02 mode. This is shown in figure 2.14.

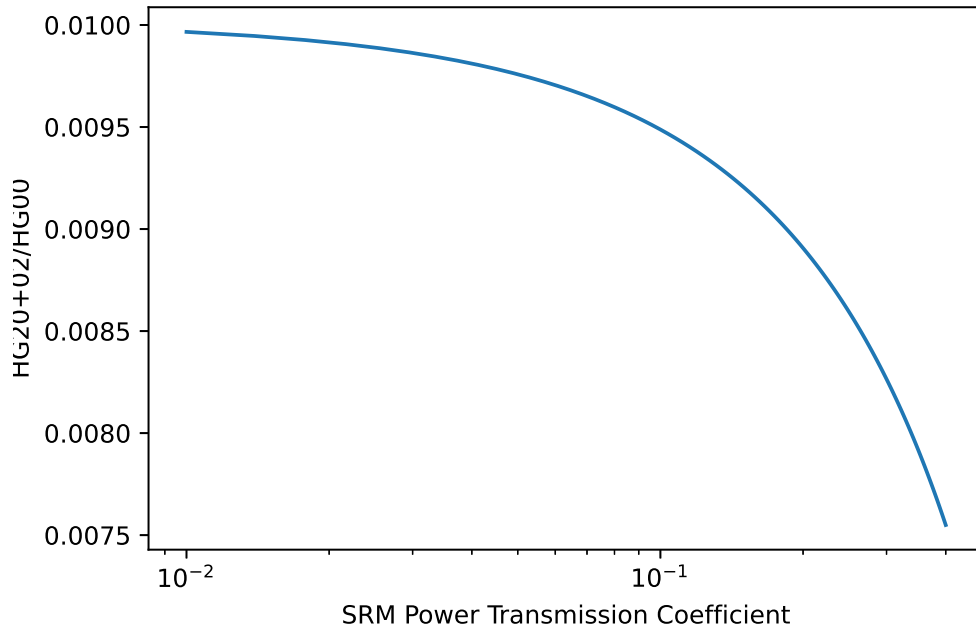


Figure 2.14: Ratio of HG20+02 to HG00 as the transmission of the SRM is increased for a mode mismatch between the arm cavity and the SRC. The interferometer is in RSE mode. A mode mismatch was introduced by setting the focal length of SR3 to be 12 mm shorter than the nominal amount.

2.6 Conclusion

Mode-mismatching between the arm cavities and the SRC results in the scattering of HG00 mode into higher order modes. Losses in HG00 usually reduces the SNR by $\sqrt{\text{Power}}$, however since there are multiple points between the squeezer and the OMC where mode-mismatches can happen, the quantum noise is enhanced by more than the usual amount for a loss. There exists a collection of measurements which show the arms are matched between 92% and 98% to the OMC.

FINESSE simulations were used to model mode-mismatches between the arm and the SRC caused by RoC errors in SR3 and SR2. The simulation used as built parameters for the SRC at the LLO site and design specification arm cavities. This allows us to study the effect of non-ideal mode-matching in a close to real life situation while reducing the complexity of the model. The interferometer was modelled in signal recycling mode and RSE mode for and locked arm cavities and single bounce configurations. It was found that there is no significant difference between single bounce and locked arm cavity configurations as there is very high levels of mode matching between the PRC and the arm cavity. In both single bounce and locked arm modes, the ITM forms a highly reflective mirror as almost all the light is reflected back. The amount of HOM is proportional to the HG00 mode in the SRC. The ratio of HOM to HG00 depends on the detuning of the SRM. As HOM do not enter the arm cavity, they cannot distinguish between the signal recycling and RSE modes of operation.

The uncertainty in the beam parameter of the arm mode at the SRM and the eigenmode of the SRC is driven by the uncertainty in SR3. This is shown in figures

2.5 and 2.6. The single pass Gouy phase of the SRC for different combinations of SR3 and SR2 was simulated, and is shown in figure 2.7. Using the measured single pass Gouy phase of the SRC, the focal length of SR2 was found to be -3.205 ± 0.003 m and the focal length of SR3 was found to be 17.971 ± 0.002 m. This is shown in figure 2.10.

Future work could involve using a heater to change the focal length of SR3. This would produce a predictable change in the focal length of SR3. This could be compared to the Gouy phase of the SRC to check that this changes the in mode matching the expected way. As the lengths between all the optics is well known, and the focal length of the ITMs is well known, the squeezer beam could be used to probe the radius of curvature of SR3 and SR2. By setting up a single bounce measurement, the incoming and outgoing beam profile could be measured and compared to a model. Having greater confidence in the beam parameter of the arm mode at the SRM would be of great benefit for the design of future upgrades to LIGO.

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